

Sets And Functions

* Defn 1.1 (i) Set theoretic difference of A and B aka complement of B relative to A

$$A \setminus B := \{x \in A \text{ st } x \notin B\}$$

(ii) If U is the universal set then $B^c := U \setminus B$

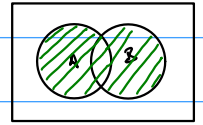
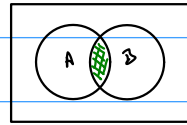
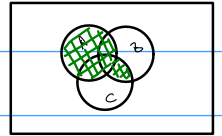
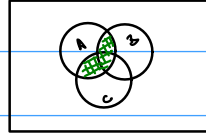
* Lemma 1.2 If A, B, C are arbitrary sets then

$$(i) A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$(ii) A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

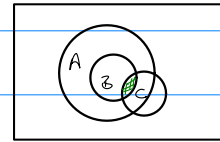
$$(iii) (A \cap B)^c = A^c \cup B^c$$

$$(iv) (A \cup B)^c = A^c \cap B^c$$



* Lemma 1.3 If A, B, C, D are sets st $D \subseteq C$ and $A \cap C = B \cap C$ then $A \cap D = B \cap D$

* Lemma 1.4 If A, B, C are sets st $B \subseteq A$ then $B \setminus C = B \cap A \setminus C$



* Lemma 1.5 If A, B, C are arbitrary sets then

$$\text{De Morgan's Laws} \quad \begin{cases} (i) A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C) \\ (ii) A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C) \\ (iii) A \setminus B = (A \setminus B) \cup (A \cap B^c) \end{cases}$$

* Hint: Use 1.2

* Caution $A \setminus (B \setminus C) \neq (A \setminus B) \setminus C$

* Defn 1.6 The set of natural numbers $\mathbb{N} := \{1, 2, 3, \dots\}$

** Well Ordering Property: Every non-empty subset of $\mathbb{N}$ has a smallest element.

* Principle of Induction
1.7

Let $P(n)$ be a statement that depends on the natural number n . Suppose that

(1) $P(1)$ is true

(2) $P(n)$ is true $\Rightarrow P(n+1)$ is true

Then $P(n)$ is true $\forall n \in \mathbb{N}$

Hint: Use Well Ordering Property of $\mathbb{N}$

* Defn 1.8

$$A \times B := \{(x, y) : x \in A, y \in B\}$$

Cartesian Product

* Defn 1.9

If A is an empty set then $A \times B = \emptyset$

* Defn 1.10

A binary relation R on a set A is a subset of $A \times A$ and it contains the pairs for whom the relation is valid.

$$aRb \text{ if } (a,b) \in R$$

* Equivalence Relation
Defn 1.11

A binary relation R on a set A s.t. $\forall a,b,c \in A$

$$(i) \quad aRa$$

$$(ii) \quad aRb \Leftrightarrow bRa$$

$$(iii) \quad aRb, bRc \Rightarrow aRc$$

* Defn 1.12

Equivalence classes are non intersecting and their union is equal to the original set

* Defn 1.13

A function $f: A \rightarrow B$ is a subset of $A \times B$ s.t. $\forall x \in A \exists$ a unique $y \in B$ s.t. $(x,y) \in f$.
 $(x,y) \in f$ is usually written as $f(x) = y$

* Defn 1.14

$$\begin{array}{cc} \text{Domain} & \text{Codomain} \\ f: \overline{A} \rightarrow \overline{B} \end{array}$$

$$\text{Range of } f \quad R(f) := \{y \in B : \exists x \in A \text{ and } y = f(x)\}$$

$$\text{Image / Direct Image of } C \subset A \quad f(C) := \{f(x) : x \in C\}$$

$$\text{Inverse Image of } D \subset B \quad f^{-1}(D) := \{x \in A : f(x) \in D\}$$

* Lemma 1.15

$f: A \rightarrow B$, $C, D \subset B$ then

$$(i) \quad f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$$

$$(ii) \quad f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$$

$$(iii) \quad f^{-1}(B \setminus C) = A \setminus f^{-1}(C)$$

* Lemma 1.16

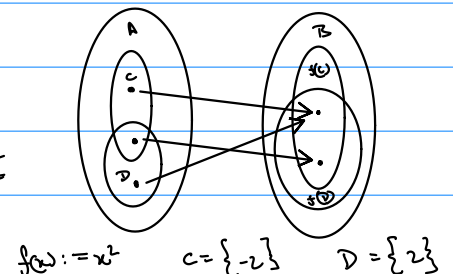
$f: A \rightarrow B$, $C, D \subset A$ then

$$(i) \quad f(C \cup D) = f(C) \cup f(D)$$

$$(ii) \quad f(C \cap D) \subset f(C) \cap f(D)$$

Proof Hint

Example



* Defn 1.17

Cardinality of A Cardinality of B

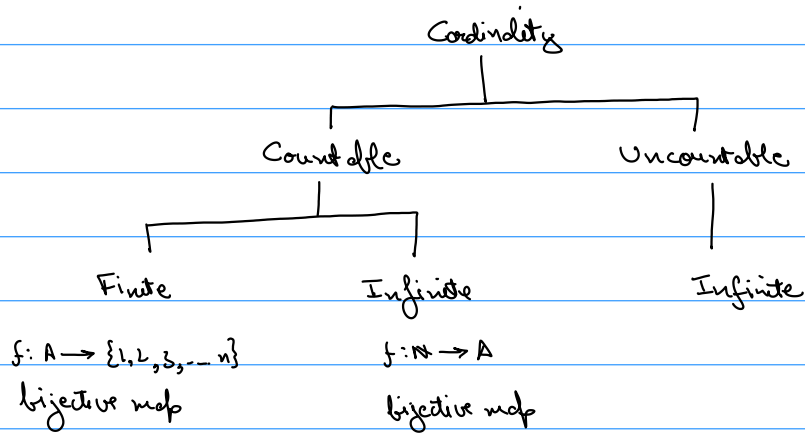
(i) $|A| = |B|$ if $\exists$ a bijective map $f: A \rightarrow B$.

(ii) $|A| \leq |B|$ " " injective " " " "

(iii) $|A| \geq |B|$ " " surjective " " " "

* Defn 1.18 The cardinality of the empty set $\emptyset$ is defined to be 0.

* Defn 1.19



* Power set
Defn 1.20

The power set $P(A)$ of a set A is defined to be the set of all subset of A

* Theorem
1.21

$|A| < |P(A)|$, rephrasing there can not exist a surjective map $f: A \rightarrow P(A)$

Proof by contradiction: Let $f: A \rightarrow P(A)$ be a surjective map

Let $S := \{x \in A : x \notin f(x)\}$

Let $c \in A$ s.t. $f(c) = S$

$\swarrow$
 $c \in S$
 $c \notin f(c)$

$\searrow$
 $c \notin S$
 $c \in f(c)$
 $c \in S$

* Lemma
1.22

If A and B are two statements that can be true or false then

$A \text{ is true} \implies B \text{ is true}$ is equivalent to $B \text{ is false} \implies A \text{ is false}$.

$\mathbb{R}$: Real Numbers

④

* Ordered set
Defn 2.1

A set S together with a binary relation $<$ s.t

- (i) $\forall x, y \in S$, exactly one is true $<, >$ or $=$
- (ii) If $x, y, z \in S$ s.t $x > y$ and $y > z$ then $x > z$

* Defn 2.2

Let $E \subseteq S$ and S be an ordered set

- (i) $b \in S$ is called an upper bound of E if $\forall x \in E \quad x \leq b$
- (ii) $b_0 \in S$ an upper bound of E is called the least upper bound or supremum if $b_0 \leq b$ where b is another upper bound
- (iii) Similarly the greatest lower bound / infimum of E is defined.

* Defn 2.3

An ordered set S has the least upper bound property if every non empty set of S that is bounded above has a least upper bound.

* Field
Defn 2.4

A set F together with two operations $(+)$ and $(\cdot)$ defined on it s.t

- ① If $x, y \in F$ then $x + y \in F$ Closure
- ② $\forall x, y, z \in F \quad (x + y) + z = x + (y + z)$ Associative
- ③ $\exists 0 \in F$ s.t $\forall x \in F \quad x + 0 = 0 + x = x$ Identity
- ④ $\forall x \in F \quad \exists -x \in F$ s.t $x + (-x) = -x + x = 0$ Inverses
- ⑤ $\forall x, y \in F \quad x + y = y + x$ Commutativity
- ⑥ If $x, y \in F$ then $x - y \in F$
- ⑦ $\forall x, y, z \in F \quad (x \cdot y) \cdot z = x \cdot (y \cdot z)$
- ⑧ $\exists 1 \in F$ s.t $\forall x \in F \quad 1 \cdot x = x \cdot 1 = x$
- ⑨ $\forall x \in \{F - \{0\}\} \quad \exists x^{-1} \in F$ s.t $x \cdot x^{-1} = x^{-1} \cdot x = 1$
- ⑩ $\forall x, y \in F \quad x \cdot y = y \cdot x$
- ⑪ $\forall x, y, z \in F \quad x \cdot (y + z) = x \cdot y + x \cdot z$ Distributive

* Defn 2.5

An ordered field F is an ordered set s.t

- (i) If $x, y, z \in F$ s.t $y < z$ then $x + y < x + z$
- (ii) If $x > 0$ and $y > 0$ then $xy > 0$

* Lemma 2.6 Let F be an ordered field and $x, y, z, w \in F$ then

- ① If $x > 0$ and $y > z$ then $xy > xz$
- ② If $x \neq 0$ then $x^2 > 0$ Hint: $-x = -x$
- ③ $1 > 0$
- ④ $0 < x < y$ then $x^2 < y^2$
- ⑤ $0 < x < y$ then $0 < 1/y < 1/x$

* Lemma 2.7 Let F be an ordered field, then

F has supremum property $\iff F$ has the infimum property

* Defn 2.8 Real Number, $\mathbb{R}$, is an unique ordered field with least upper bound property s.t. $\mathbb{Q} \subset \mathbb{R}$.

* Lemma 2.9 If $S \subset \mathbb{R}$ is nonempty and upper bounded, then $\forall \epsilon > 0 \exists x \in S$ s.t.
 $\sup(S) - \epsilon < x \leq \sup(S)$

Hint: Proof by contradiction

* Lemma 2.10 Let $A, B \subset \mathbb{R}$ be nonempty subsets s.t. $x \leq y \forall x \in A$ and $y \in B$, then
 $\sup(A) \leq \inf(B)$

* Lemma 2.10.1 Let $A, B, C \subset \mathbb{R}$ s.t. (i) $|A| = |B|$ (ii) $A = \{a_1, a_2, \dots\}$, $B = \{b_1, b_2, \dots\}$ and
 $C = \{a_1 + b_1, a_2 + b_2, \dots\}$

- (i) If A and B are bounded above $\implies \sup(C) = \sup(A) + \sup(B)$
- (ii) If A and B are bounded below $\implies \inf(C) = \inf(A) + \inf(B)$

* Archimedian Property of $\mathbb{R}$ 2.11 If $x, y \in \mathbb{R}$ s.t. $x > 0$ and $x < y$ then $\exists n \in \mathbb{N}$ s.t. $nx > y$.

Hint: Rephrase $\mathbb{N} \subset \mathbb{R}$ is unbounded above and use Contradiction with 2.9

* Theorem 2.2

$\mathbb{Q}$ is dense in $\mathbb{R}$ i.e. If $x, y \in \mathbb{R}$ s.t. $x < y$ then

$$\exists z \in \mathbb{Q} \text{ s.t. } x < z < y$$

Proof hint: Let $x > 0$ $\exists n \in \mathbb{N}$ s.t. $n(y-x) > 1$. $A := \{k : k \in \mathbb{N} \text{ and } k > nx\}$

$m \in A$ be the smallest element of A . Then $m > nx \Rightarrow x < m/n$

$$\text{Also } m-1 < nx \Rightarrow m < nx+1 \Rightarrow m < ny \Rightarrow m/n < y$$

* Defn 2.3

Let $A \subset \mathbb{R}$ and $x \in \mathbb{R}$

$$x+A = \{x+y \in \mathbb{R} : y \in A\}$$

$$xA = \{xy \in \mathbb{R} : y \in A\}$$

* Lemma 2.14

If $A \subset \mathbb{R}$ is bounded above, then $\sup(x+A) = x + \sup(A)$

If $x > 0$, and A is bounded above $\sup(xA) = x \sup(A)$

* Defn 2.15

Let $A \subset \mathbb{R}$

(i) if A is empty then $\sup(A) := -\infty$
 $\inf(A) := \infty$

(ii) If A is not bounded above then $\sup(A) := \infty$

(iii) " " " " below then $\inf(A) := -\infty$

* Defn 2.16

When $A \subset \mathbb{R}$ is bounded above s.t. $\sup(A) \in A$ then $\sup(A)$ is denoted by $\max(A)$. Similarly $\min(A)$ is defined

* Defn 2.17

Absolute value function $f: \mathbb{R} \rightarrow [0, \infty)$

$$|x| := \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

* Lemma 2.18

$$(i) |x| = 0 \iff x = 0$$

$$(ii) |xy| = |x| |y|$$

$$(iii) |x| \leq y \iff -y \leq x \leq y$$

$$(iv) -|x| \leq x \leq |x| \quad \forall x \in \mathbb{R}$$

Hint: Divide it into different cases based on signs of x and y

* Lemma 2.19 (Triangle Inequalities)

- (i) $|x+y| \leq |x|+|y| \quad \forall x, y \in \mathbb{R}$
- (ii) $|x-y| \leq |x|+|y| \quad "$
- (iii) $||x|-|y|| \leq |x-y| \quad "$

* Defn 2.20 Let $f: D \rightarrow \mathbb{R}$. We say f is bounded above if $\exists M \in \mathbb{R}$ s.t.

$$|f(x)| \leq M \quad \forall x \in D$$

* Lemma 2.21 If $f: D \rightarrow \mathbb{R}$ and $g: D \rightarrow \mathbb{R}$ are bounded functions s.t.

$$f(x) \leq g(x) \quad \forall x \in D \text{ then}$$

$$\sup_{x \in D} f(x) \leq \sup_{x \in D} g(x) \quad \inf_{x \in D} f(x) \leq \inf_{x \in D} g(x)$$

* Defn 2.22 Intervals

$$[a, b] := \{x \in \mathbb{R} \text{ s.t. } a \leq x \leq b\}$$

$$(a, b) := \{x \in \mathbb{R} \text{ s.t. } a < x < b\}$$

* Lemma 2.23 All intervals have the same cardinality i.e. $\exists$ a bijective mdp b/w any two intervals of $\mathbb{R}$

Example $(-\pi/2, \pi/2) \xrightarrow{\tan(x)} \underbrace{(-\infty, \infty)}_{\mathbb{R}}$

* Theorem 2.24 $\mathbb{R}$ is uncountable

Proof: Construct an element $y \in \mathbb{R}$ s.t. y is not an element of countably infinite set.

Let $x \in \mathbb{R}$ s.t. (i) x is countably infinite (ii) x is dense in $\mathbb{R}$

Inductive construction of A

$$\left\{ \begin{array}{l} \text{let } a_1 := x_1 \quad b_1 := x_1 + 1 \\ a_1 < a_2 < \dots < a_{n-1} < b_{n-1} < \dots < b_2 < b_1 \text{ s.t.} \\ x_j \in (a_k, b_k) \quad \forall 1 \leq j \leq n-1 \text{ and } 1 \leq k \leq n-1 \\ \text{find smallest } x_i \in A \text{ s.t. } x_i \in (a_n, b_n) \quad a_n := x_i \quad \text{take } b_n \in (a_n, b_{n-1}) \end{array} \right.$$

$$A = \{a_1, a_2, \dots, a_n\}, \quad B = \{b_1, b_2, \dots, b_n\}$$

Let $y = \sup(B) \in \mathbb{R}$ s.t. $y \notin A$, $y \in B$ and $y \neq x$

$\therefore \mathbb{R}$ is uncountable

* Defn 2.25 For an infinite sequence of digits $0.d_1d_2d_3\dots$

$$D_n := \frac{d_1}{10} + \frac{d_2}{10^2} + \dots + \frac{d_n}{10^n}$$

* Lemma 2.26 (i) Every infinite sequence of digits $0.d_1d_2\dots$ represents unique $x \in [0,1]$ st

$$D_n \leq x \leq D_n + \frac{1}{10^n} \quad \forall n \in \mathbb{N}$$

(ii) $\forall x \in (0,1]$ $\exists$ a unique infinite sequence of digits $0.d_1d_2d_3\dots$ that represents x st

$$D_n \leq x \leq D_n + \frac{1}{10^n} \quad \forall n \in \mathbb{N}$$

Sequences And Series

9

* Defn 3.1 A sequence of real numbers, i.e. $\{x_n\}_{n=1}^{\infty}$ is a function $f: \mathbb{N} \rightarrow \mathbb{R}$

* Defn 3.2 $\{x_n\}_{n=1}^{\infty}$ is called bounded if the underlying function is bounded.
Defn 2.20

* Defn 3.3 $\{x_n\}_{n=1}^{\infty}$ is said to converge to $x \in \mathbb{R}$, i.e. $\lim_{n \rightarrow \infty} x_n = x$ if $\forall \epsilon > 0 \exists M \in \mathbb{N}$ s.t. $\forall n > M \quad |x_n - x| < \epsilon$

* Defn 3.4 A sequence that is not convergent is called divergent.

* Lemma 3.5 A convergent sequence has a unique limit.

* Lemma 3.6 A convergent sequence is bounded.

Hint: $|x_n - x| < |x_n - x| + |x| < \epsilon + |x|$

* Lemma 3.7 If $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ are convergent sequences s.t. $x_n \leq y_n \quad \forall n \in \mathbb{N}$ then $\lim_{n \rightarrow \infty} x_n \leq \lim_{n \rightarrow \infty} y_n$

Hint $\exists M \in \mathbb{N}$ s.t. $n > M$
 $x_n - y_n < \epsilon/2$
 $y_n > x_n - \epsilon/2$

* Defn 3.8 $\{x_n\}_{n=1}^{\infty}$ is called monotone increasing if $x_{n+1} \geq x_n \quad \forall n \in \mathbb{N}$
" " " " decreasing " " $x_{n+1} \leq x_n$ " "

* Theorem 3.9 A monotone sequence is bounded if and only if it is convergent. (Hint use 2.9)
if $\{x_n\}_{n=1}^{\infty}$ is monotone increasing and bounded $\lim_{n \rightarrow \infty} x_n = \sup \{x_n : n \in \mathbb{N}\}$
" " " " decreasing " " $\lim_{n \rightarrow \infty} x_n = \inf \{x_n : n \in \mathbb{N}\}$

* Lemma 3.10 For $\{x_n\}_{n=1}^{\infty}$ the following statements are equivalent:

(i) $\{x_n\}_{n=1}^{\infty}$ is convergent

(ii) The k tail, i.e. $\{x_n\}_{n=k}^{\infty}$ converges $\forall k \in \mathbb{N}$

(iii) " " " " " converges for some $k \in \mathbb{N}$

* Defn 3.4

$\{x_{n_i}\}_{i=1}^{\infty}$ is called a subsequence of $\{x_n\}_{n=1}^{\infty}$ if $\{n_i\}_{i=1}^{\infty}$ is a strictly increasing sequence of natural numbers.

* Lemma 3.2

If $\{x_n\}_{n=1}^{\infty}$ is a convergent sequence then all its subsequences $\{x_{n_i}\}_{i=1}^{\infty}$ are convergent to the same limit

* Squeeze lemma:

3.13

If $\{x_n\}_{n=1}^{\infty}$, $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are sequences and $L \in \mathbb{R}$ s.t

(i) $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are convergent sequences

$$(ii) \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = L$$

$$(iii) a_n \leq x_n \leq b_n \quad \forall n \in \mathbb{N}$$

then $\{x_n\}_{n=1}^{\infty}$ also converges to L

* Lemma 3.14

Let $\{x_n\}_{n=1}^{\infty}$, $\{y_n\}_{n=1}^{\infty}$ be convergent sequences then

$$(i) \{z_n\}_{n=1}^{\infty} \text{ where } z_n = x_n + y_n \text{ converges to } \lim_{n \rightarrow \infty} z_n = \lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n$$

$$(ii) \{z_n\}_{n=1}^{\infty} \text{ where } z_n = x_n y_n \text{ converges to } \lim_{n \rightarrow \infty} z_n = \lim_{n \rightarrow \infty} x_n \lim_{n \rightarrow \infty} y_n$$

Proof Hint: Let $\varepsilon > 0$. $K = \max(|x_1|, |y_1|, \frac{1}{2}, 1)$

$$\exists M_1 \in \mathbb{N} \text{ s.t } \forall n > M_1, |x_n - x| < \varepsilon/3K$$

$$\exists M_2 \in \mathbb{N} \text{ s.t } \forall n > M_2, |y_n - y| < \varepsilon/3K$$

$$\forall n > \max(M_1, M_2) \text{ Prove } |x_n y_n - xy| < \varepsilon$$

so that $\{x_n\}$ is a sequence of real numbers

so that $\{1/y_n\}_{n=1}^{\infty}$ is bounded

(ii) if $y_n \neq 0 \quad \forall n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} y_n \neq 0$ then $\{z_n\}_{n=1}^{\infty}$ where $z_n = x_n / y_n$ converges to $\lim_{n \rightarrow \infty} z_n = \lim_{n \rightarrow \infty} x_n / \lim_{n \rightarrow \infty} y_n$

Proof Hint: Prove $\{1/y_n\}_{n=1}^{\infty}$ converges to $1/y$ where $y = \lim_{n \rightarrow \infty} y_n$

$$\text{Let } \varepsilon > 0 \quad \exists M \in \mathbb{N} \text{ s.t } |y_n - y| < \min\left(|y| \frac{\varepsilon}{2}, \frac{|y|}{2}\right), \quad \forall n > M \quad \frac{1}{|y_n|} < \frac{2}{|y|}$$

$$\text{Prove } \left| \frac{1}{y_n} - \frac{1}{y} \right| < \varepsilon$$

**

If $\{x_n\}_{n=1}^{\infty}$, $\{y_n\}_{n=1}^{\infty}$ are convergent sequences s.t $y_n \neq 0 \quad \forall n \in \mathbb{N}$, then even with $\lim_{n \rightarrow \infty} y_n = 0$ it is possible for $\{x_n/y_n\}_{n=1}^{\infty}$ to be convergent. Example $x_n = \frac{1}{n^2}$ $y_n = \frac{1}{n}$

* Lemma 3.15 If $\{x_n\}_{n=1}^{\infty}$ is a convergent sequence s.t. $x_n \geq 0 \quad \forall n \in \mathbb{N}$, then

$$(i) \quad \lim_{n \rightarrow \infty} \sqrt{x_n} = \sqrt{\lim_{n \rightarrow \infty} x_n}$$

$$(ii) \quad \lim_{n \rightarrow \infty} |x_n| = \left| \lim_{n \rightarrow \infty} x_n \right|$$

* Lemma 3.16 Let $\{x_n\}_{n=1}^{\infty}$ and $\{a_n\}_{n=1}^{\infty}$ be sequences and $x \in \mathbb{R}$ s.t.

$$(i) \quad \lim_{n \rightarrow \infty} a_n = 0$$

$$(ii) \quad |x_n - x| < a_n \quad \forall n \in \mathbb{N}$$

then $\{x_n\}_{n=1}^{\infty}$ is convergent and $\lim_{n \rightarrow \infty} x_n = x$

* Lemma 3.17 Let $\{x_n\}_{n=1}^{\infty}$ be a sequence s.t. (i) $x_n \neq 0 \quad \forall n \in \mathbb{N}$

$$(ii) \quad L = \lim_{n \rightarrow \infty} \frac{|x_{n+1}|}{|x_n|} \text{ exists}$$

then (a) if $L < 1$ then $\lim_{n \rightarrow \infty} x_n = 0$

Hint: $L < 1 \quad \exists M \in \mathbb{N} \quad \forall n \geq M$

(b) if $L > 1$ then $\{x_n\}_{n=1}^{\infty}$ is unbounded

$$\frac{|x_{n+1}|}{|x_n|} < L < 2 \quad \forall n \geq M$$

$$|x_n| = 2^{n-M} |x_M|$$

* Corollary 3.18 If $|a| < 1$ the $\{a^n\}_{n=1}^{\infty}$ converges to 0

* Defn 3.19 Let $\{x_n\}_{n=1}^{\infty}$ be a bounded sequence. Define $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ sequences

$$\text{where } a_n := \sup \{x_k : k \geq n\} \quad b_n := \inf \{x_k : k \geq n\}$$

$$\text{Limit superior } \limsup_{n \rightarrow \infty} x_n := \lim_{n \rightarrow \infty} a_n = \inf \{a_n : n \in \mathbb{N}\}$$

$$\text{Limit inferior } \liminf_{n \rightarrow \infty} x_n := \lim_{n \rightarrow \infty} b_n = \sup \{b_n : n \in \mathbb{N}\}$$

* Lemma 3.20 If $\{x_n\}_{n=1}^{\infty}$ is a bounded sequence then $\liminf_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} x_n$

Using 3.7

* Theorem 3.21 If $\{x_n\}_{n=1}^{\infty}$ is a bounded sequence, then $\exists$ subsequences $\{x_{n_i}\}_{i=1}^{\infty}, \{x_{m_j}\}_{j=1}^{\infty}$ s.t.

$$\limsup_{n \rightarrow \infty} x_n = \lim_{i \rightarrow \infty} x_{n_i}$$

Need not to be monotone

$$\liminf_{n \rightarrow \infty} x_n = \lim_{j \rightarrow \infty} x_{m_j}$$

Proof:
 Inductive construction: $\left\{ \begin{array}{l} \text{let } n_1 = 1 \\ \text{let } n_1, n_2, \dots, n_{k-1} \text{ are already defined s.t. } |x_{n_i} - a_{n_i}| < \frac{1}{i} \quad 1 \leq i \leq k-1 \\ \text{let } m \in \mathbb{N} \text{ s.t. } m \geq n_{k-1} \text{ and } |a_{n_{k-1}+1} - x_m| < \frac{1}{k} \text{ Define } n_k = m \\ \text{then } \forall k \geq 2 \quad |a_{n_k} - x_{n_k}| < \frac{1}{k} \text{ as } a_{n_k} \leq a_{n_{k-1}+1} \end{array} \right.$

Using 3.2 and 3.19 let $a = \lim_{n \rightarrow \infty} a_n = \lim_{k \rightarrow \infty} a_{n_k}$

Let $\epsilon > 0$, find $M \in \mathbb{N}$ s.t. $\forall k \geq M$ ① $k \geq 2$ ② $\frac{1}{k} < \epsilon/2$ ③ $|a - a_{n_k}| < \epsilon/2$

then $|x_{n_k} - a| \leq |x_{n_k} - a_{n_k}| + |a_{n_k} - a| < \frac{1}{k} + \epsilon/2 < \epsilon/2 + \epsilon/2 < \epsilon$

* Theorem 3.22

let $\{x_n\}_{n=1}^{\infty}$ be a bounded sequence. Then

Proof Hint: Use 3.21, 3.12, 3.13

$$\lim_{n \rightarrow \infty} x_n = x \iff \liminf_{n \rightarrow \infty} x_n = \limsup_{n \rightarrow \infty} x_n = x$$

* Lemma 3.23

A bounded sequence $\{x_n\}_{n=1}^{\infty}$ is convergent to x if and only if every convergent subsequence $\{x_{n_k}\}_{k=1}^{\infty}$ converges to x

Use 3.22 and 3.13

* Corollary 3.24

For a convergent sequence all the subsequences converge to same limit but we prefer the structure of lemma 3.23 as it reduces the number of subsequences to be tested.

* Theorem 3.25

Bolzano-Weierstrass: Suppose a sequence $\{x_n\}_{n=1}^{\infty}$ of real numbers is bounded. Then there exists a convergent subsequence $\{x_{n_k}\}_{k=1}^{\infty}$.

Proof:

Inductive definition

Let $a_1, b_1 \in \mathbb{R}$ s.t. $a_1 \leq x_n \leq b_1$

Let's say $a_1, a_2, \dots, a_{k-1}, b_{k-1}, \dots, b_1$ s.t.

① $a_1 \leq a_2 \leq a_3 \leq \dots \leq a_{k-1} \leq b_{k-1} \leq \dots \leq b_1$

② There are no number of points b/w $[a_i, b_i]$ $1 \leq i \leq k-1$

③ $x_{n_i} \in (a_i, b_i)$ $1 \leq i \leq k-1$

let $c_k = (b_{k-1} + a_{k-1})/2$

if ∞ # of points in $[b_{k-1}, c_k]$ take $b_k = b_{k-1}$ $a_k = c_k$

" " " " " " $[c_k, a_k]$ take $b_k = c_k$ $a_k = a_{k-1}$

choose $x_{n_k} \in \{x_n\}_{n=1}^{\infty}$ s.t. $a_k \leq x_{n_k} \leq b_k$

Defn $\{x_n\}_{n=1}^{\infty}$ diverges to infinity, i.e. $\lim_{n \rightarrow \infty} x_n = \infty$ if $\forall K \in \mathbb{R} \exists M \in \mathbb{N}$ s.t.

$$\forall n \geq M \quad x_n > K$$

lemma 3.29 If a sequence is Cauchy, then it is bounded. Hint: Similar to 3.6

Proof Hint Cauchy $\Rightarrow$ bounded $\Rightarrow \exists \{x_{n_i}\}_{i=1}^{\infty}$ s.t. $\lim_{i \rightarrow \infty} x_{n_i} = \limsup_{n \rightarrow \infty} x_n = a$
 $\downarrow$ $\exists \{x_{m_j}\}_{j=1}^{\infty}$ s.t. $\lim_{j \rightarrow \infty} x_{m_j} = \liminf_{n \rightarrow \infty} x_n = b$

$$\begin{aligned} \exists M_1 \in \mathbb{N} \text{ st } \forall i \geq M_1 \quad |x_{n_i} - a| &< \varepsilon/3 & |a - b| &= |a - x_{n_i} + x_{n_i} - x_{m_i} + x_{m_i} - b| \\ \exists M_2 \in \mathbb{N} \text{ " } \forall j \geq M_2 \quad |x_{m_j} - b| &< \varepsilon/3 & &\leq |x_{n_i} - a| + |x_{n_i} - b| + |x_{n_i} - x_{m_j}| \\ \exists M_3 \in \mathbb{N} \text{ " } \forall i \geq M_3 \quad |x_{n_i} - x_{m_j}| &< \varepsilon/3 & &< \varepsilon/3 + \varepsilon/3 + \varepsilon/3 < \varepsilon \\ & & &\therefore a = b \\ & & &\text{Use 3.22} \end{aligned}$$

* Defn 3.31 Given a sequence $\{x_n\}_{n=1}^{\infty}$ of real numbers, a series is defined as $\sum_{n=1}^{\infty} x_n$.

A series is called convergent if the sequence of its partial sums $\{s_k\}$ is convergent.
where $s_k := \sum_{n=1}^k x_n = x_1 + x_2 + \dots + x_k$

* Defn 3.32 A series $\sum_{n=1}^{\infty} x_n$ is called Cauchy if the sequence of partial sums $\{s_k\}$ is Cauchy.

* Lemma 3.33 The series $\sum_{n=1}^{\infty} x_n$ is Cauchy if and only if $\forall \epsilon > 0 \exists \text{ an } M \in \mathbb{N} \text{ s.t.}$
 $\forall n \geq M \text{ and every } k > n$
 $|\sum_{i=n+1}^k x_i| < \epsilon$

* Lemma 3.34 $\sum_{n=1}^{\infty} x_n$ is a convergent series $\implies \lim_{n \rightarrow \infty} x_n = 0$

Hint: Use 3.33

* Lemma 3.35 Let $\alpha \in \mathbb{R}$ and $\sum_{n=1}^{\infty} x_n, \sum_{n=1}^{\infty} y_n$ be convergent series then

(i) $\sum_{n=1}^{\infty} \alpha x_n$ is also a convergent series and further

$$\sum_{n=1}^{\infty} \alpha x_n = \alpha \sum_{n=1}^{\infty} x_n$$

(ii) $\sum_{n=1}^{\infty} (x_n + y_n)$ is a convergent series and further

$$\sum_{n=1}^{\infty} (x_n + y_n) = \sum_{n=1}^{\infty} x_n + \sum_{n=1}^{\infty} y_n$$

* * Caution: Because $\sum_{n=1}^k x_n y_n \neq \sum_{n=1}^k x_n \sum_{n=1}^k y_n$ for arbitrary x_n, y_n and $n \geq 2$ we can not say if $\sum_{n=1}^{\infty} (x_n y_n)$ is convergent or not.

* Defn 3.36 A series $\sum_{n=1}^{\infty} x_n$ is called absolutely convergent if the series $\sum_{n=1}^{\infty} |x_n|$ converges

* Defn 3.37 A series is called conditionally convergent if it is convergent but not absolutely convergent.

* Lemma 3.38 If a series $\sum_{n=1}^{\infty} x_n$ is absolutely convergent then it is also convergent Use 3.33

* Lemma 3.39 Let $\sum_{n=1}^{\infty} x_n$ and $\sum_{n=1}^{\infty} y_n$ are series s.t. $\overbrace{0 \leq x_n \leq y_n}^{\text{imp}} \quad \forall n \in \mathbb{N}$ then
 (comparison test)
 (i) If $\sum_{n=1}^{\infty} y_n$ is convergent then $\sum_{n=1}^{\infty} x_n$ is also convergent
 (ii) If $\sum_{n=1}^{\infty} x_n$ diverges to ∞ then $\sum_{n=1}^{\infty} y_n$ also diverges to ∞ .

* Lemma 3.40 Let $\sum_{n=1}^{\infty} x_n$ is a series s.t. $x_n \neq 0 \quad \forall n \in \mathbb{N}$ and if $L = \lim_{n \rightarrow \infty} \frac{|x_{n+1}|}{|x_n|}$ then
 (Ratio test)

- (i) If $L < 1$ then $\sum_{n=1}^{\infty} x_n$ converges absolutely Handy Hint: Use 3.17 and 3.18
 (ii) If $L > 1$ then $\sum_{n=1}^{\infty} x_n$ diverges Use 3.34

Continuous Functions

(16)

* Cluster Point 4.1 $x \in \mathbb{R}$ is called a cluster point of $S \subset \mathbb{R}$ if $\forall \varepsilon > 0 \exists y \in S \setminus \{x\}$ s.t. $|x-y| < \varepsilon$
i.e. the set $(x-\varepsilon, x+\varepsilon) \cap S \setminus \{x\}$ is non empty

* Lemma 4.2 $x \in \mathbb{R}$ is a cluster point of $S \subset \mathbb{R}$ $\iff \exists \{x_n\}_{n=1}^{\infty}$ s.t. $\forall n \ x_n \in S \setminus \{x\}$ and $\lim_{n \rightarrow \infty} x_n = x$

* Limit of a function 4.3 Let $c \in \mathbb{R}$ be a cluster point of $S \subset \mathbb{R}$, $f: S \rightarrow \mathbb{R}$ and $L \in \mathbb{R}$ then
We say $\lim_{x \rightarrow c} f(x) = L$ if $\forall \varepsilon > 0, \exists \delta > 0$ s.t.
if $x \in S \setminus \{c\}$ and $|x-c| < \delta \Rightarrow |f(x)-L| < \varepsilon$

* Lemma 4.4 If $\lim_{x \rightarrow c} f(x)$ exists then the limit is unique

* Lemma 4.5 (Sequential Limit) Let $c \in \mathbb{R}$ be a cluster point of $S \subset \mathbb{R}$, $f: S \rightarrow \mathbb{R}$ and $L \in \mathbb{R}$ then
 $\lim_{x \rightarrow c} f(x) = L \iff \forall \{x_n\}_{n=1}^{\infty}$ s.t. $\forall n \in \mathbb{N} \ x_n \in S \setminus \{c\}$ and $\lim_{n \rightarrow \infty} x_n = c$ and $\{f(x_n)\}_{n=1}^{\infty}$ converges to L .

Proof hint: If A is false B is false
 $\exists$ an $\varepsilon > 0$ s.t. $\forall \delta > 0 \exists x \in S \setminus \{c\}$ s.t. $|x-c| < \delta$ and $|f(x)-L| \geq \varepsilon$

* Corollary 4.6 Let $c \in \mathbb{R}$ be a cluster point of $S \subset \mathbb{R}$, $f: S \rightarrow \mathbb{R}$ and $g: S \rightarrow \mathbb{R}$ are functions s.t.
 $f(x) \leq g(x) \ \forall x \in S$, and $\exists \lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ then
 $\lim_{x \rightarrow c} f(x) \leq \lim_{x \rightarrow c} g(x)$ Proof hint: Use 4.5, 3.7

* Squeeze lemma 4.7 Let $c \in \mathbb{R}$ be a cluster point of $S \subset \mathbb{R}$, $f: S \rightarrow \mathbb{R}$, $g: S \rightarrow \mathbb{R}$, $h: S \rightarrow \mathbb{R}$ functions s.t.
(i) $g(x) \leq f(x) \leq h(x) \ \forall x \in S$
(ii) $\exists \lim_{x \rightarrow c} g(x)$ and $\lim_{x \rightarrow c} h(x)$
(iii) $\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x)$ then

$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x)$ Proof hint: Use 4.5, 3.3

* Corollary 4.8

Let $c \in \mathbb{R}$ a cluster point of $S \subset \mathbb{R}$, $f: S \rightarrow \mathbb{R}$, $g: S \rightarrow \mathbb{R}$ are functions s.t $\exists \lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$, then

$$(i) \quad \lim_{x \rightarrow c} (f(x) + g(x)) = \left(\lim_{x \rightarrow c} f(x) \right) + \left(\lim_{x \rightarrow c} g(x) \right)$$

$$(ii) \quad \lim_{x \rightarrow c} (f(x) \cdot g(x)) = \left(\lim_{x \rightarrow c} f(x) \right) \left(\lim_{x \rightarrow c} g(x) \right)$$

Proof Hint: 4.5, 3.14

(iii) if $g(x) \neq 0 \quad \forall x \in S \setminus \{c\}$ and $\lim_{x \rightarrow c} g(x) \neq 0$ then

$$\lim_{x \rightarrow c} \left(\frac{f(x)}{g(x)} \right) = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)}$$

* It is possible that $\lim_{x \rightarrow c} g(x) = 0$ and $\lim_{x \rightarrow c} \left(\frac{f(x)}{g(x)} \right)$ exists but we can't be sure

* Restriction of f to A
4.10

Let $f: S \rightarrow \mathbb{R}$ and $A \subset S$. The function $f|_A: A \rightarrow \mathbb{R}$ is given by

$$f|_A(x) := f(x) \quad \text{for } x \in A$$

* Lemma 4.11

Let $S \subset \mathbb{R}$, $c \in \mathbb{R}$, $f: S \rightarrow \mathbb{R}$, $A \subset S$ and $\exists \alpha > 0$ s.t

$$(A \setminus \{c\}) \cap (c - \alpha, c + \alpha) = (S \setminus \{c\}) \cap (c - \alpha, c + \alpha)$$

(i) c is a cluster point of $A \iff c$ is a cluster point of S

Hint $A \Leftarrow B$

Let $\epsilon > 0$ $\delta = \min(\epsilon, \alpha)$

$\exists x \in S \setminus \{c\}$ $0 < \delta \leq \alpha$

$x \in S \setminus \{c\} \cap (c - \alpha, c + \alpha)$

$x \in A \setminus \{c\} \cap (c - \alpha, c + \alpha)$

(ii) Suppose c is a cluster point of S then

$$\lim_{x \rightarrow c} f(x) = L \iff \lim_{x \rightarrow c} f|_A(x) = L$$

* Defn 4.12

Let $f: S \rightarrow \mathbb{R}$, $c \in \mathbb{R}$ a cluster point of $S \cap (c, \infty)$ and $\exists \lim_{x \rightarrow c} f|_{S \cap (c, \infty)}(x)$ then

(One sided limit)

$$\lim_{x \rightarrow c^+} f(x) := \lim_{x \rightarrow c} f|_{S \cap (c, \infty)}(x)$$

* Lemma 4.13

Let $S \subset \mathbb{R}$, $f: S \rightarrow \mathbb{R}$, $c \in \mathbb{R}$ a cluster point of $S \cap (c, \infty)$ and $S \cap (-\infty, c)$ then

$$\lim_{x \rightarrow c} f(x) = L \iff \lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^-} f(x) = L$$

* Continuous
function 4.14

Let $S \subset \mathbb{R}$, $c \in S$ and $f: S \rightarrow \mathbb{R}$. Then f is called continuous at c if $\forall \epsilon > 0 \exists \delta > 0$
s.t if $x \in S$ and $|x - c| < \delta$ then $|f(x) - f(c)| < \epsilon$

* Defn 4.15

$f: S \rightarrow \mathbb{R}$ is called continuous if it is continuous at $\forall c \in S$

* Lemma
4.16

Let $S \subset \mathbb{R}$, $f: S \rightarrow \mathbb{R}$, $c \in S$. then

(i) if c is not a cluster point of S then f is continuous at c .

(ii) if c is a cluster point of S , then

$$f \text{ is continuous at } c \iff \lim_{x \rightarrow c} f(x) = f(c)$$

$$(iii) f \text{ is continuous at } c \iff \forall \{x_n\}_{n=1}^{\infty} \text{ s.t. } \forall n \in \mathbb{N} \quad \underline{x_n \in S} \text{ and } \lim_{n \rightarrow \infty} x_n = c, \\ \{f(x_n)\}_{n=1}^{\infty} \text{ converges to } f(c)$$

Hint: Same
logic as 4.5

* Lemma
4.17

Let $f: S \rightarrow \mathbb{R}$ and $g: S \rightarrow \mathbb{R}$ be continuous functions at $c \in S$. then

(i) $h: S \rightarrow \mathbb{R}$ $h(x) := f(x) + g(x)$ is continuous at c

Use 4.16 with
3.14

(ii) " " " $h(x) := f(x) \cdot g(x)$ " " " " " "

(iii) If $g(x) \neq 0 \forall x \in S$ then $h: S \rightarrow \mathbb{R}$ $h(x) := f(x)/g(x)$ is continuous at c .

Note $\lim_{x \rightarrow c} g(x) \neq 0$ is not needed and will not even exist if c is not a cluster point

* Lemma
4.18

Let $A, B \subset \mathbb{R}$, $f: B \rightarrow \mathbb{R}$, $g: A \rightarrow B$ be functions s.t g and f are continuous at c and $g(c)$ respectively

then $f \circ g: A \rightarrow \mathbb{R}$ where $f \circ g(x) := f(g(x))$ is continuous at c .

Use 4.16

* Caution

The closedness of interval is an integral feature for the following lemma's and theorems.

* Lemma 4.19

$f: [a, b] \rightarrow \mathbb{R}$ is continuous $\Rightarrow f: [a, b] \rightarrow \mathbb{R}$ is bounded.

Hint: Contrapositive statement (1.23)

+

Bolzano's Weierstrass Theorem

+

4.16(ii)

* Extreme value theorem
4.20

A continuous function $f: [a, b] \rightarrow \mathbb{R}$ on a closed and bounded interval $[a, b]$ achieves both an absolute maximum and an absolute minimum on $[a, b]$

Hint: Use 4.19, 2.9 to construct $\{f(x_n)\}_{n=1}^{\infty}$ s.t. $\lim_{n \rightarrow \infty} f(x_n) = \sup(f([a, b]))$

Bolzano Weierstrass $\{x_n\}_{n=1}^{\infty}$ s.t. $\lim_{n \rightarrow \infty} x_n = x \in [a, b]$, then $f(x) = \sup(f([a, b]))$

* Lemma
4.21

Let $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function. Suppose $f(a) < 0$ and $f(b) > 0$, then $\exists c \in (a, b)$ s.t. $f(c) = 0$

Proof: Similar to Bolzano's Weierstrass construct $\{a_n\}_{n=1}^{\infty}, \{b_n\}_{n=1}^{\infty}$ s.t.

$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = x \in [a, b]$ and $f(a_n) < 0$, $f(b_n) > 0$

Using 4.16 (iii) $f(x) = 0$

* Intermediate Value theorem
4.22

Let $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function. Suppose $y \in \mathbb{R}$ is s.t.

$f(a) < y < f(b)$ or $f(a) > y > f(b)$. Then $\exists c \in (a, b)$ s.t. $f(c) = y$
Hint: $g(x) := f(x) - y$

* Corollary
4.23 (i)

If $f: [a, b] \rightarrow \mathbb{R}$ is continuous and non constant function and $f(x_0) > 0$ where $x_0 \in (a, b)$ then

$\exists$ interval I s.t. (i) $x_0 \in I$ (ii) $f(I)$ is an interval

(iii) $f(x) > 0 \forall x \in I$

* Corollary
4.23 (ii)

If $f: [a, b] \rightarrow \mathbb{R}$ is a non constant function, f is continuous at x_0 and $f(x_0) > 0$ where

$x_0 \in (a, b)$ then $\exists$ interval I s.t. (i) $x_0 \in I$ (ii) $f(x) > 0 \forall x \in I$

Example $f(x) := \begin{cases} 0 & \text{if } x \text{ is rational} \\ x & \text{if } x \text{ is irrational} \end{cases}$

$\lim_{x \rightarrow 0} f(x) = 0 \Rightarrow f$ is continuous at $x = 0$

* Defn
4.24

Let $S \subset \mathbb{R}$, $f: S \rightarrow \mathbb{R}$. f is called uniformly continuous if $\forall \varepsilon > 0 \exists \delta > 0$ s.t.

if $x, y \in S$ s.t. $|x - y| < \delta$ then $|f(x) - f(y)| < \varepsilon$.

* Theorem
4.25

$f: [a, b] \rightarrow \mathbb{R}$ is continuous function

$\implies$

$f: [a, b] \rightarrow \mathbb{R}$ is uniformly continuous

Hint:- Use contrapositive statement to construct $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ st (i) $|x_n - y_n| < \frac{1}{n}$ where $n \in \mathbb{N}$
(ii) $|f(x_n) - f(y_n)| \geq \epsilon$. The using Bolzano Weierstrass construct $\{x_{n_i}\}_{i=1}^{\infty}$ st $\lim_{i \rightarrow \infty} x_{n_i} = L$.
Show $\lim_{i \rightarrow \infty} y_{n_i} = L$. Then show either $\{f(x_{n_i})\}_{i=1}^{\infty}$ or $\{f(y_{n_i})\}_{i=1}^{\infty}$ doesn't converge to $f(c)$ and use 4.16(i).

* Lemma 4.26 Let $f: S \rightarrow \mathbb{R}$ be a uniformly continuous function and $\{x_n\}_{n=1}^{\infty}$ where $\forall n \in \mathbb{N} \ x_n \in S$ is a Cauchy sequence then $\{f(x_n)\}_{n=1}^{\infty}$ is also Cauchy

* Lemma 4.27 $f: (a, b) \rightarrow \mathbb{R}$ is uniformly continuous $\iff \exists L = \lim_{x \rightarrow a^+} f(x)$, and $\tilde{f}(x) := \begin{cases} f(x) & \text{if } x \in (a, b) \\ L & \text{if } x = a \\ L & \text{if } x = b \end{cases}$ is continuous
 $L = \lim_{x \rightarrow b^-} f(x)$
Hint Use 4.26 and 4.5 to show L exists

* Defn 4.28 $f: S \rightarrow \mathbb{R}$ is called Lipschitz continuous, if $\exists K \in \mathbb{R}$ s.t.
 $|f(x) - f(y)| < K|x - y|$

* Lemma 4.29 A Lipschitz continuous function is uniformly continuous.
Hint for $\epsilon > 0$ use $\delta = \epsilon/K$

* Defn 4.30 We say ∞ is a cluster point of SCR if $\forall M \in \mathbb{R} \ \exists x \in S$ s.t. $x \geq M$

* Defn 4.31 Let ∞ be a cluster point of SCR, $f: S \rightarrow \mathbb{R}$ a function, $L \in \mathbb{R}$. We say $\lim_{x \rightarrow \infty} f(x) = L$ if
 $\forall \epsilon > 0 \ \exists M \in \mathbb{R}$ s.t. if $x \geq M$ then $|f(x) - L| < \epsilon$

* Lemma 4.32 Let ∞ be a cluster point SCR, $f: S \rightarrow \mathbb{R}$ a function, then
 $\lim_{x \rightarrow \infty} f(x) = L \iff \lim_{n \rightarrow \infty} f(x_n) = L$ where $\{x_n\}_{n=1}^{\infty}$ is a sequence st $\forall n \in \mathbb{N} \ x_n \in S, \lim_{n \rightarrow \infty} x_n = \infty$

* Defn 4.32 Let ∞ be a cluster point SCR, $f: S \rightarrow \mathbb{R}$ a function. We say $\lim_{x \rightarrow \infty} f(x) = \infty$ if
 $\forall N \in \mathbb{R} \ \exists M \in \mathbb{R}$ s.t. if $x \geq M$ then $f(x) > N$

* Composition of limits 4.33 Suppose $f: A \rightarrow B, g: B \rightarrow \mathbb{R}, A, B \subset \mathbb{R}, a, b, c \in \mathbb{R} \cup \{\infty, -\infty\}$, a, b are cluster points of A and B respectively, $\lim_{x \rightarrow a} f(x) = b, \lim_{y \rightarrow b} g(y) = c$ and
if $b \in B$ then $g(b) = c$ then $\lim_{x \rightarrow a} g(f(x)) = c$

* lemma
4.34

Let $S \subset \mathbb{R}$, $c \in \mathbb{R}$, $f: S \rightarrow \mathbb{R}$ be a monotone increasing function,

(i) If c is a cluster point of $S \cap (-\infty, c) \Rightarrow \lim_{x \rightarrow c^-} f(x) = \sup \{f(x) : x < c, x \in S\} = a$
 Hint: Consider $a = \infty$ only possible if $\exists x \in S$ s.t. $x > c$
 $a \in \mathbb{R}$

(ii) If ∞ is a cluster point of S

$$\lim_{x \rightarrow \infty} f(x) = \sup \{f(x) : x \in S\}$$

(iii) if $-\infty$ is a cluster point of S

$$\lim_{x \rightarrow -\infty} f(x) = \inf \{f(x) : x \in S\}$$

* lemma
4.35

If $f: S \rightarrow \mathbb{R}$ is strictly increasing monotone function, $c \in S$ is an interior cluster point,

$$\lim_{x \rightarrow c^-} f(x) \leq \lim_{x \rightarrow c^+} f(x)$$

* lemma
4.36

Interval
 $I \subset \mathbb{R}$, $f: I \rightarrow \mathbb{R}$ is monotone and not constant

$f(I)$ is an interval $\iff f$ is continuous

Proof Hint: Intermediate Value theorem and Cauchy's criterion statement with 4.35

* lemma
4.37

$I \subset \mathbb{R}$, $f: I \rightarrow \mathbb{R}$ is monotone $\Rightarrow$ Then f has countably many discontinuities

Proof Hint: Use 4.36

* lemma
4.38

$I \subset \mathbb{R}$, $f: I \rightarrow \mathbb{R}$ is strictly monotone, then $f^{\pm}: \underbrace{f(I)}_{\text{range}} \rightarrow \mathbb{R}$ is continuous

Proof Hint: let $y \in f(I)$
 $\begin{cases} \rightarrow \text{Noncluster point (4.16i)} \\ \rightarrow \text{cluster point} \end{cases}$

$y_0 \in \mathbb{R}$ be a interior cluster point of I , $\exists$ unique x_0 s.t. $f(x_0) = y_0$

Using 4.29 $\lim_{y \rightarrow y_0^+} f^+(y) = \inf \{f^+(y) : y \geq y_0\} = \inf \{x : x \geq x_0\} = x_0$

$$\lim_{y \rightarrow y_0^-} f^-(y) = \sup \{f^-(y) : y \leq y_0\} = \sup \{x : x \leq x_0\} = x_0$$

$$\therefore \lim_{y \rightarrow y_0^+} f^+(y) = \lim_{y \rightarrow y_0^-} f^-(y) = f^-(y_0)$$

Derivatives

22

Interval

* Defn 5.1 Let $f: I \rightarrow \mathbb{R}$, we say f is differentiable at $c \in I$ if the following limit exists

$$f'(c) := \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

* Lemma 5.2 If $f: I \rightarrow \mathbb{R}$ is differentiable at $c \in I \Rightarrow f$ is continuous at c

Hint Use 4.8 and 4.16 (ii)

* Lemma 5.3 Let $f: I \rightarrow \mathbb{R}$, $g: I \rightarrow \mathbb{R}$ be differentiable at $c \in I$, then

(i) $h: I \rightarrow \mathbb{R}$ $h(x) := \alpha f(x)$ where $\alpha \in \mathbb{R}$, $h'(c) = \alpha f'(c)$

(ii) $h: I \rightarrow \mathbb{R}$ $h(x) := f(x) + g(x)$ $h'(c) = f'(c) + g'(c)$

(iii) $h: I \rightarrow \mathbb{R}$ $h(x) := f(x)g(x)$ $h'(c) = f'(c)g(c) + f(c)g'(c)$

(iv) if $g(x) \neq 0 \forall x \in I$ then $h: I \rightarrow \mathbb{R}$ $h(x) := \frac{f(x)}{g(x)}$ $h'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}$

* Chain Rule Lemma 5.4 Let $g: I_1 \rightarrow I_2$ be differentiable at $c \in I_1$ and $f: I_2 \rightarrow \mathbb{R}$ be differentiable at $g(c) \in I_2$ then $h: I_1 \rightarrow \mathbb{R}$ $h(x) := f(g(x))$ is differentiable at c and $h'(c) = f'(g(c))g'(c)$

Hint Continuous 4.16 (ii)
$$\overline{u(x)} := \begin{cases} \frac{g(x) - g(c)}{x - c} & x \neq c \\ g'(c) & x = c \end{cases}$$
 Continuous
$$\overline{u(y)} := \begin{cases} \frac{f(y) - f(a)}{y - a} & y \neq a \\ f'(a) & y = a \end{cases}$$

Let $y = g(x)$ and $x \neq c$ then $u(g(x)) \overline{u(x)} = \frac{f(g(x)) - f(g(c))}{x - c}$ Use 4.8 (ii)

* Relative maxima 5.5 Let $S \subset \mathbb{R}$ and $f: S \rightarrow \mathbb{R}$. f is said to have relative maxima at $c \in S$ if $\exists \delta > 0$ st

$$\text{if } |x - c| < \delta \Rightarrow f(x) \leq f(c)$$

* Lemma 5.6 Let $f: (a, b) \rightarrow \mathbb{R}$ be differentiable at $c \in (a, b)$ and c is a relative maxima/minima then $f'(c) = 0$

Proof Let $f(x)$ has relative minima at $x = c$

Construct $\{x_n\}_{n=1}^{\infty}$ st $x_n \leq c \forall n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} x_n = c$ then $\frac{f(x_n) - f(c)}{x_n - c} \leq 0$

Similarly $\{y_n\}_{n=1}^{\infty}$ st $y_n \geq c \forall n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} y_n = c$ then $\frac{f(y_n) - f(c)}{y_n - c} \geq 0$

Using 4.16
$$f'(c) = \lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c} = \lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c}$$

Using 4.5 $f'(c) = 0$

* Lemma 5.7
(Rolle's theorem)

Let $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function, differentiable on (a, b) s.t. $f(a) = f(b)$
then $\exists c \in (a, b)$ s.t. $f'(c) = 0$ Hint: Use 5.6

* Mean Value theorem 5.8

Let $f: [a, b] \rightarrow \mathbb{R}$ be a continuous function, differentiable on (a, b) . Then $\exists c \in (a, b)$
s.t. $f'(c) = \frac{f(b) - f(a)}{b - a}$ Hint: Use 5.7
 $g(x) = f(x) - \left(\frac{f(b) - f(a)}{b - a}\right)(x - a)$

* Lemma 5.9

$f: I \rightarrow \mathbb{R}$ is a differentiable function s.t. $f'(x) = 0 \quad \forall x \in I \iff f$ is a constant function

* Lemma 5.10

Let $f: I \rightarrow \mathbb{R}$ be a differentiable function

(i) f is increasing $\iff f'(x) \geq 0 \quad \forall x \in I$ Hint: Use Mean Value Theorem

(ii) f is decreasing $\iff f'(x) \leq 0 \quad \forall x \in I$

* Darboux theorem 5.11

Let $f: [a, b] \rightarrow \mathbb{R}$ be a differentiable function, suppose $y \in \mathbb{R}$ s.t. $f'(a) < y < f'(b)$ or $f'(a) > y > f'(b)$ then $\exists c \in (a, b)$ s.t. $f'(c) = y$.

Hint: $g(x) := yx - f(x)$, $g'(a) < 0$ $g'(b) > 0$ or $g'(a) > 0$ $g'(b) < 0$, use 5.6

* Corollary 5.12

Let $h(x) := f'(x)$ where $f: [a, b] \rightarrow \mathbb{R}$ is a differentiable function then using Darboux theorem $h(x)$ cannot have a jump discontinuity

* Defn 5.14

If $f: I \rightarrow \mathbb{R}$ is differentiable and f' is continuous on I , then f is called continuously differentiable

* Caution

Let $f: I \rightarrow \mathbb{R}$ and $x_0 \in I$

(i) If f is twice differentiable at x_0 we need f' to exist over an interval J s.t. $x_0 \in J$ and using 5.2 f will be continuous over J

(ii) If f is only differentiable once at x_0 there might not any interval J s.t. $x_0 \in J$ and f is continuous over J

* Defn 5.15

For a n times differentiable function f defined near a point $x_0 \in \mathbb{R}$, the n^{th} order

Taylor polynomial for f at x_0 is

$$P_n^{x_0}(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)(x-x_0)^2}{2!} + \dots + \frac{f^{(n)}(x_0)(x-x_0)^n}{n!}$$

* Theorem 5.14
Taylor

Suppose for $f: [a, b] \rightarrow \mathbb{R}$ $\exists$ n continuous derivatives on $[a, b]$ and $(n+1)$ derivative on (a, b) . Then given distinct $x_0, x \in [a, b]$, $\exists c \in (x_0, x)$ s.t.

$$f(x) = P_n^{x_0}(x) + \frac{f^{(n+1)}(c)}{(n+1)!} (x-x_0)^{n+1}$$

Hint: Let $g(x) = f(x) - \left[P_n^{x_0}(x) + \frac{M(x-x_0)^{n+1}}{(n+1)!} \right]$, Compute M s.t. $g(x) = 0$, Then use mean value Theorem recursively

* Caution
5.16.1

If f is infinitely divisible then $f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x-x_0)^k$.

There no guarantee that this series converges to $f(x)$ for any $x \neq x_0$.

Even where it converge there is no guarantee that it converges to f .

Functions f whose Taylor series is convergent to f at every point within an open interval containing x_0 are called analytical functions

* Lemma
5.17

Let $I, J \subset \mathbb{R}$ be intervals, $f: I \rightarrow J$ be a strictly monotone onto function.

Then if f is differentiable at $x_0 \in I$ and $f'(x_0) \neq 0$, then f^{-1} is differentiable at

$$y_0 = f(x_0) \text{ and } (f^{-1})'(y_0) = \frac{1}{f'(f^{-1}(y_0))} = \frac{1}{f'(x_0)}$$

Hint: Use 4.35,

Hint: Let $f(x_0) = y_0$, $f(x) = y$ then if $y \neq y_0$ $\frac{g(y) - g(y_0)}{y - y_0} = \frac{x - x_0}{f(x) - f(x_0)}$

Let $u(y) := \begin{cases} \frac{x - x_0}{f(x) - f(x_0)} & x \neq x_0 \\ \frac{1}{f'(x_0)} & x = x_0 \end{cases}$, then using 4.18 $u(g(y))$ is continuous at y_0 .

$$\text{Using 4.16(ii)} \quad \lim_{y \rightarrow y_0} u(g(y)) = u(g(y_0)) = (f^{-1})'(y_0) = \frac{1}{f'(g(y_0))}$$

* Inverse
Function
Theorem
5.18

Let $f: (a, b) \rightarrow \mathbb{R}$ be a continuously differentiable function, $x_0 \in (a, b)$ where $f'(x_0) \neq 0$.

Then $\exists$ an open interval $I \subset (a, b)$ s.t. $x_0 \in I$ and $f|_I$ is injective with a continuously differentiable inverse $g: J \rightarrow I$ where $J := f(I)$ and

$$g'(y) = \frac{1}{f'(g(y))} \quad \forall y \in J$$

Integrals

25

* Defn 6.1 A Partition P of $[a, b]$ is a finite set of numbers $\{x_0, x_1, \dots, x_n\}$ s.t.
 $a = x_0 < x_1 < x_2 < \dots < x_n = b$

* Defn 6.2 Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded function and P a partition of $[a, b]$
 $m_i := \inf \{f(x) : x_{i-1} \leq x \leq x_i\}$ $M_i := \sup \{f(x) : x_{i-1} \leq x \leq x_i\}$
 $L(P, f) := \sum_{i=1}^n m_i \Delta x_i$ $U(P, f) := \sum_{i=1}^n M_i \Delta x_i$
 lower darbour sum upper darbour sum

* Defn 6.3 Let $P = \{x_0, x_1, \dots, x_n\}$ and $\tilde{P} = \{\tilde{x}_0, \tilde{x}_1, \dots, \tilde{x}_r\}$ be partitions of $[a, b]$.
 $\tilde{P}$ is called a refinement of P if $P \subset \tilde{P}$ when treated as a set

* Lemma 6.4 Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded function, P a partition of $[a, b]$ and $\tilde{P}$ a refinement of P ,
 then $L(P, f) \leq L(\tilde{P}, f) \leq U(\tilde{P}, f) \leq U(P, f)$

* Defn 6.5 Lower Darbour Integral $:= \int_a^b f(x) dx := \sup \{L(P, f) : P \text{ is a partition of } [a, b]\}$
 Upper " " $:= \int_a^b f(x) dx := \inf \{U(P, f) : P \text{ is a partition of } [a, b]\}$

* Lemma 6.6 Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded function then $\int_a^b f(x) dx \leq \int_a^b f(x) dx$
 Hint $P = P_1 \cup P_2$
 6.5 $L(P_1, f) \leq U(P_2, f)$
 Use 2.10

* Riemann integrable 6.7 function Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded function s.t $\int_a^b f(x) dx = \int_a^b f(x) dx$ then f is called Riemann integrable and $\int_a^b f(x) dx := \int_a^b f(x) dx = \int_a^b f(x) dx$

* Lemma 6.8 $\star$ Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded function. Then if $\forall \epsilon > 0 \exists$ a partition P of $[a, b]$ s.t
 $U(f, P) - L(f, P) < \epsilon$ then f is Riemann integrable
 Hint: $\int_a^b f - \int_a^b f \leq U(f, P) - L(f, P)$

* Defn 6.9 Let $f: [a, b] \rightarrow \mathbb{R}$ be a Riemann integrable function of $[a, b]$ then
 $\int_a^a f := - \int_a^b f$ and $\int_a^b f := 0$ $y \in [a, b]$

* Lemma 6.10

Let $a < b < c$ and $f: [a, c] \rightarrow \mathbb{R}$ be a bounded function. Then

$$\int_a^c f = \int_a^b f + \int_b^c f \quad \text{and} \quad \int_a^c f = \int_a^b f + \int_b^c f$$

Hint: 6.4 $P = P_1 \cup P_2$ ^{partition of $[b, c]$}
 Partition of $[a, b]$

* Lemma 6.11

Let $a < b < c$

$f: [a, c] \rightarrow \mathbb{R}$ is Riemann integrable $\iff$ $f: [a, b] \rightarrow \mathbb{R}$ and $f: [b, c] \rightarrow \mathbb{R}$ are Riemann integrable

further $\int_a^c f = \int_a^b f + \int_b^c f$

Hint 6.10

* Lemma 6.12

If $f: [a, b] \rightarrow \mathbb{R}$ and $g: [a, b] \rightarrow \mathbb{R}$ are bounded functions then

$$\int_a^b (f+g) \geq \int_a^b f + \int_a^b g \quad \int_a^b (f+g) \leq \int_a^b f + \int_a^b g$$

Hint $L(P, f+g) \geq L(P, f) + L(P, g)$
 $P = P_1 \cup P_2$
 6.5 and 2.10.1

* Lemma 6.13

Let $f, g \in \mathcal{R}([a, b])$ and $\alpha \in \mathbb{R}$

(i) $\alpha f \in \mathcal{R}([a, b])$ and $\int_a^b \alpha f = \alpha \int_a^b f$

Hint $\alpha > 0$
 $\alpha < 0$ (cases)

(ii) $f+g \in \mathcal{R}([a, b])$ and $\int_a^b f+g = \int_a^b f + \int_a^b g$

Use 6.12

* Theorem 6.14

If $f: [a, b] \rightarrow \mathbb{R}$ is continuous $\implies f \in \mathcal{R}([a, b])$

Hint Continuous $\xrightarrow{4.25}$ Uniformly continuous

$\exists \delta > 0$ s.t. if $|x-y| < \delta$
 $|f(x) - f(y)| < \epsilon_{\frac{1}{b-a}}$

Use 6.8

* Lemma 6.15

Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded function, $\{a_n\}_{n=1}^\infty, \{b_n\}_{n=1}^\infty$ be sequences s.t.

$$a < \underline{a_n} < b_1 < b, \quad \lim_{n \rightarrow \infty} a_n = a, \quad \lim_{n \rightarrow \infty} b_n = b \quad \text{and} \quad f \in \mathcal{R}([a_n, b_n])$$

then $f \in \mathcal{R}([a, b])$ and $\int_a^b f = \lim_{n \rightarrow \infty} \int_{a_n}^{b_n} f$

Hint: Use Bolzano Weierstrass
 to find $\lim_{k \rightarrow \infty} \int_{a_{n_k}}^{b_{n_k}} f dx = L$

Show $L = \int_a^b f = \int_a^b f \leq L$ Use 3.24

* Theorem 6.16

$f: [a, b] \rightarrow \mathbb{R}$ be a bounded function with finitely many discontinuities then f is Riemann integrable.

Use 6.15

* Lemma 6.17

Let $f: [a, b] \rightarrow \mathbb{R} \in \mathcal{R}([a, b])$, $g: [a, b] \rightarrow \mathbb{R}$ s.t. $f(x) = g(x) \quad \forall x \in [a, b] \setminus S$, where S is finite then $g \in \mathcal{R}([a, b])$ and $\int_a^b g = \int_a^b f$

* Lemma 6.18

Let $f: [a, b] \rightarrow \mathbb{R}$ be a monotone function. Then $f \in \mathcal{R}([a, b])$

Hint: let $\Delta x_i = \frac{b-a}{n}$
 $m_i = f(x_{i-1}), M_i = f(x_i)$, Use 6.8

* Fundamental Theorem of Calculus 6.19
 First form

Let $F: [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Let $f \in \mathcal{R}([a, b])$ s.t. $f(x) = F'(x) \quad \forall x \in (a, b)$ then

$$\int_a^b f(x) dx = F(b) - F(a)$$

Hint: $f \in \mathcal{R}([a, b])$ $\leftarrow$ Bounded
 $\int_a^b f = \int_a^b f$
 Use Mean Value theorem

* Fundamental Theorem of Calculus 6.20
 Second form

Let $f \in \mathcal{R}([a, b])$, then $F(x) := \int_a^x f(u) du$ is continuous on $[a, b]$ and

if f is continuous at $c \in [a, b]$ then F is differentiable at c and $F'(c) = f(c)$

Hint: Prove $F(x)$ is
 (i) Lipschitz continuous
 (ii) $f(x) - \epsilon \leq f(x) \leq f(x) + \epsilon$
 $(x_1 - \epsilon)(x - c) \leq \int_c^x f(u) du \leq (f(x) + \epsilon)(x - c)$

* Change of Variables Theorem 6.21

Let $g: [a, b] \rightarrow \mathbb{R}$ be a continuously differentiable function, $f: [c, d] \rightarrow \mathbb{R}$ be a continuous function, and suppose $g([a, b]) \subset [c, d]$. Then

$$\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(s) ds$$

Proof: g is continuously differentiable $\rightarrow g'$ is continuous. as f is continuous as well

$f \circ g$ is continuous on $[a, b] \therefore \in \mathcal{R}([a, b])$

$$F(x) := \int_{g(a)}^x f(s) ds \quad \text{then} \quad F(g(a)) = 0$$

Since $f \in \mathcal{R}([a, b])$ and continuous using 6.20 $F(x)$ is differentiable and $F'(x) = f(x)$

Since $g(x)$ is also differentiable, $F \circ g(x)$ is differentiable and $(F \circ g)'(x) = F'(g(x)) g'(x) = f(g(x)) g'(x)$

$$\int_a^b f(g(x)) g'(x) dx = F(g(b)) - F(g(a)) = \int_a^b F'(g(x)) dx = \int_a^b f(g(x)) g'(x) dx$$

* Defn 6.22
Log

There exists a unique function $L: (0, \infty) \rightarrow \mathbb{R}$ s.t.

$$\left. \begin{array}{l} \text{(i)} \quad L(1) = 0 \\ \text{(ii)} \quad L'(x) = \frac{1}{x} \end{array} \right\} \rightarrow L(x) := \int_1^x \frac{1}{s} ds$$

Hint: Uniqueness follows using first form of the fundamental theorem of calculus

* Properties of
log 6.23

$$\begin{array}{l} \text{(i)} \quad L(xy) = L(x) + L(y) \\ \text{(ii)} \quad L \text{ is strictly increasing, bijective and} \end{array}$$

$$\lim_{x \rightarrow 0} L(x) = -\infty \quad \text{and} \quad \lim_{x \rightarrow \infty} L(x) = \infty$$

$$\text{(iii)} \quad \text{If } q \in \mathbb{Q} \quad L(x^q) = q L(x)$$

$$\text{Hint } g(x) := L(xy) - L(x) \Rightarrow g'(x) = 0, g(1) = L(y)$$

$$\text{Let } y > 0 \quad L(z^y) = y L(z) \quad \text{Use arithmetic properties and intermediate value theorem}$$

properties and intermediate value theorem

$$0 = L(x/x) = L(x) + L(1/x) = L(1/x) = -L(x)$$

$$L(x) = L(x^{1/n}) = \frac{1}{n} L(x^n)$$

* Defn 6.24
exponential

There exists a unique function $E: \mathbb{R} \rightarrow (0, \infty)$ s.t.

$$\left. \begin{array}{l} \text{(i)} \quad E(0) = 1 \\ \text{(ii)} \quad E'(x) = E(x) \end{array} \right\} \rightarrow E := \text{Inverse function of } L$$

Hint: inverse function theorem 5.18

uniqueness let $F(x)$ and $G(x)$ be two

$$\text{solutions, } (F(x)E(-x))' = 0 \quad F(x)E(-x) = F(0)E(0) = 1$$

$$E(x-x) = E(x)E(-x) = 1 \quad \downarrow \quad F(x)E(-x) - E(x)E(-x) = 0$$

* Properties of
Exp 6.25

$$\text{(i)} \quad E \text{ is strictly increasing, bijective and}$$

$$\lim_{x \rightarrow -\infty} E(x) = 0, \quad \lim_{x \rightarrow \infty} E(x) = \infty$$

$$\text{(ii)} \quad E(x+y) = E(x)E(y)$$

$$\text{Hint } E(x+y) = E(L(x) + L(y)) = E(L(xy)) = xy = E(x)E(y)$$

$$\text{(iii)} \quad \text{If } q \in \mathbb{Q} \quad E(x^q) = (E(x))^q$$

* Defn 6.26

$$\text{If } x > 0, y \in \mathbb{R} \quad x^y := \exp(y \ln(x))$$

* Note 6.27

If y is irrational x^y can be approximated using $\{x^{y_n}\}_{n=1}^{\infty}$ where $y_n \in \mathbb{Q}$ $x \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} y_n = y$ because $\exp$ is a continuous function of y .

Sequence of Functions

(29)

* Defn 7.1
Pointwise Convergence
Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of functions $f_n: S \rightarrow \mathbb{R}$. We say $\{f_n\}_{n=1}^{\infty}$ converges pointwise to $f: S \rightarrow \mathbb{R}$ if $\forall x \in S$ and $\forall \epsilon > 0 \exists M \in \mathbb{N}$ s.t. $n \geq M \implies |f_n(x) - f(x)| < \epsilon$.

* Defn 7.2
Uniform Convergence
Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of functions $f_n: S \rightarrow \mathbb{R}$. We say $\{f_n\}_{n=1}^{\infty}$ converges uniformly to $f: S \rightarrow \mathbb{R}$ if $\forall \epsilon > 0 \exists M \in \mathbb{N}$ s.t. if $n \geq M$ then $|f_n(x) - f(x)| < \epsilon \quad \forall x \in S$.

* Defn 7.3
Uniform norm
Let $f: S \rightarrow \mathbb{R}$ be a bounded function.
$$\|f\|_S := \sup\{|f(x)| : x \in S\}$$

* Lemma 7.4
Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of bounded functions $f_n: S \rightarrow \mathbb{R}$, then
 $\{f_n\}_{n=1}^{\infty}$ converges uniformly to $f: S \rightarrow \mathbb{R} \iff \lim_{n \rightarrow \infty} \|f_n - f\|_S = 0$

* Defn 7.5
Uniformly Cauchy
Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of bounded functions $f_n: S \rightarrow \mathbb{R}$. We say $\{f_n\}_{n=1}^{\infty}$ is uniformly Cauchy or Cauchy in uniform norm if $\forall \epsilon > 0 \exists M \in \mathbb{N}$ s.t. $n, k \geq M \implies \|f_n - f_k\|_S < \epsilon$

* Lemma 7.6
Let $f_n: S \rightarrow \mathbb{R}$ be bounded functions.
 $\{f_n\}_{n=1}^{\infty}$ is Cauchy in uniform norm $\iff \exists f: S \rightarrow \mathbb{R}$ s.t. $\{f_n\}_{n=1}^{\infty}$ converges uniformly to f .

Hint $A \Rightarrow B$ Define f using pointwise convergence and 3.30

* Theorem 7.7
Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of continuous functions $f_n: S \rightarrow \mathbb{R}$ that converges uniformly to $f: S \rightarrow \mathbb{R} \implies f$ is continuous.

Hint: Let $x \in S$, $\{x_j\}_{j=1}^{\infty}$ s.t. $x_j \in S \quad \forall j \in \mathbb{N}$ and $\lim_{j \rightarrow \infty} x_j = x$ then

Using 7.2 find $k \in \mathbb{N}$ s.t. $|f_k(y) - f(y)| < \epsilon/3 \quad \forall y \in S$

Using 4.16 find $M \in \mathbb{N}$ s.t. $j \geq M \implies |f_k(x_j) - f_k(x)| < \epsilon/3$

Prove that for $j \geq M \implies |f(x_j) - f(x)| < \epsilon$

* Theorem 7.8
 Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of Riemann integrable functions $f_n: [a, b] \rightarrow \mathbb{R}$. If $\{f_n\}_{n=1}^{\infty}$ converges uniformly to $f: I \rightarrow \mathbb{R} \implies f \in R([a, b])$ and $\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \int_a^b f_n(x) dx$

Hint Use triangle Inequality to prove f is bounded

Use 6.12 with $M \in \mathbb{N}$ s.t. if $k \geq M$ $|f_k(x) - f(x)| < \frac{\epsilon}{2(b-a)}$ to show
 $\int_a^b f - \int_a^b f_k < \epsilon$

* Theorem 7.9
 (i) Let I be a bounded interval
 (ii) $f_n: I \rightarrow \mathbb{R}$ be continuously differentiable $\implies \{f_n\}_{n=1}^{\infty}$ converges uniformly to a continuously differentiable function $f: I \rightarrow \mathbb{R}$ and $f' = g$
 (iii) $\{f'_n\}$ converges uniformly to $g: I \rightarrow \mathbb{R}$
 (iv) $\exists c \in I$ s.t. $\{f_n(c)\}_{n=1}^{\infty}$ is a convergent sequence

Hint $f_n(x) = f_n(c) + \int_c^x f'_n(t) dt$

$$f(x) := \lim_{n \rightarrow \infty} \left[f_n(c) + \int_c^x f'_n(t) dt \right] = f(c) + \int_c^x g(t) dt$$

Let b, a are upper and lower bounds of I . Find $M \in \mathbb{N}$ s.t. $\forall n \geq M$ $|f'_n(x) - g(x)| < \frac{\epsilon}{b-a} \quad \forall x \in I$

$$\forall n \geq M \quad |f_n(x) - f(x)| = \left| \int_c^x (f'_n(t) - g(t)) dt \right| < \frac{\epsilon}{b-a} (b-a) < \epsilon$$

* Defn 7.10
 Continuity
 Let $\mathcal{U} \subset \mathbb{R}^2$ be a set and $F: \mathcal{U} \rightarrow \mathbb{R}$ be a function. Then F is called continuous at $(x, y) \in \mathcal{U}$ if $\forall \{x_n, y_n\}_{n=1}^{\infty} \subset \mathcal{U}$ s.t. $\lim_{n \rightarrow \infty} x_n = x$ and $\lim_{n \rightarrow \infty} y_n = y$ we have
 $\lim_{n \rightarrow \infty} F(x_n, y_n) = F(x, y)$

* Piccard's Theorem 7.11
 Let $I, J \subset \mathbb{R}$ be bounded and closed intervals, $(x_0, y_0) \in I_0 \times J_0$ where I_0, J_0 are interiors of I and J .
 $F: I \times J \rightarrow \mathbb{R}$ s.t. F is continuous and Lipschitz continuous in second variable i.e.
 $\exists L \in \mathbb{R}$ s.t. $\forall y, z \in J, x \in I \quad |F(x, y) - F(x, z)| \leq L|y - z|$, then
 $\exists h > 0$ s.t. $[x_0 - h, x_0 + h] \subset I$ and there exists a unique differentiable function
 $f: [x_0 - h, x_0 + h] \rightarrow J$ s.t. $f'(x) = F(x, f(x))$ and $f(x_0) = y_0$.

Proof
 (i) Since F is a continuous function on a closed bounded interval it is bounded, i.e.
 $\exists M \in \mathbb{R}$ s.t. $|F(x, y)| \leq M \quad \forall (x, y) \in I \times J$

(ii) without loss of generality let $x_0 = 0$

choose $\alpha > 0$ st $[-\alpha, \alpha] \subset I$ and $[y_0 - \alpha, y_0 + \alpha] \subset J$

$$\text{let } h = \min\left(\alpha, \frac{\alpha}{M+L\alpha}\right)$$

Lipschitz constant

(iii) Inductively define $\{f_n\}_{n=1}^{\infty}$ s.t. (a) $f_n: [-h, h] \rightarrow [y_0 - \alpha, y_0 + \alpha]$

(b) f_n is continuous

$f_0(x) = y_0$, Assume we have defined $f_0, f_1, \dots, f_{k-1}$ s.t they satisfy (a) and (b)

$$f_k(x) := y_0 + \int_0^x F(y, f_{k-1}(t)) dt$$

continuous continuous

(c) Show $f_k: [-h, h] \rightarrow [y_0 - \alpha, y_0 + \alpha]$

$$|f_k(x) - y_0| = \left| \int_0^x F(y, f_{k-1}(t)) dt \right| \leq Mh \leq \frac{M\alpha}{M+L\alpha} < \alpha$$

(iv) Show $\{f_n\}_{n=1}^{\infty}$ is uniformly Cauchy $\Rightarrow$ uniformly convergent

$$\begin{aligned} \text{Let } n, k \in \mathbb{N}, \quad n > k, \quad |f_n(x) - f_k(x)| &= \left| \int_0^x (F(t, f_{n-1}(t)) - F(t, f_{k-1}(t))) dt \right| \leq \left| \int_0^x L(f_{n-1}(t) - f_{k-1}(t)) dt \right| \\ &\leq \frac{L\alpha}{M+L\alpha} \|f_{n-1} - f_{k-1}\|_{[-h, h]} \leq \frac{C}{<1} \|f_{n-1} - f_{k-1}\|_{[-h, h]} \\ &\leq C^k \|f_{n-k} - f_0\| \\ &\leq C^k \|f_{n-k} - y_0\| \leq C^k \alpha \end{aligned}$$

$\therefore \{f_n\}_{n=1}^{\infty}$ is uniformly Cauchy $\xRightarrow{7.6}$ uniformly convergent. $\lim_{n \rightarrow \infty} f_n = f$.

(v) Show f is the solution of the equation.

Since f_n are continuous and the convergence is uniform $\Rightarrow f$ is continuous

$$|F(t, f_n(t)) - F(t, f(t))| \leq L|f_n(t) - f(t)| \leq L\|f_n - f\|_{[-h, h]}$$

$\therefore F(t, f_n(t))$ converges uniformly to $F(t, f(t))$

$$\begin{aligned} y_0 + \int_0^x F(t, f(t)) dt &= y_0 + \int_0^x F(t, \lim_{n \rightarrow \infty} f_n(t)) dt, \text{ since } F \text{ is continuous} \\ &= y_0 + \int_0^x \lim_{n \rightarrow \infty} F(t, f_n(t)) dt, \text{ since } F \text{ converges uniformly} \\ &= \lim_{n \rightarrow \infty} \left(y_0 + \int_0^x F(t, f_n(t)) dt \right) \\ &= \lim_{n \rightarrow \infty} f_{n+1}(x) = f(x) \end{aligned}$$

(vi) Solution is unique. let $f(x)$ and $g(x)$ be two solutions then $|f(x) - g(x)| \leq \left| \int_0^x F(t, f(t)) - F(t, g(t)) dt \right|$

$$|f(x) - g(x)| \leq \frac{C}{<1} \|f - g\|_{[-h, h]} \quad \text{taking sup } \|f - g\|_{[-h, h]} = 0$$

Metric Spaces

37

* Defn 8.1

Metric Space

Let X be a set and $d: X \times X \rightarrow \mathbb{R}$ be a function s.t. $\forall x, y, z \in X$

$$(i) \quad d(x, y) \geq 0$$

$$(ii) \quad d(x, y) = 0 \iff x = y$$

$$(iii) \quad d(x, y) = d(y, x)$$

$$(iv) \quad d(x, z) \leq d(x, y) + d(y, z)$$

* Defn 8.2

If (X, d) is a metric space, $Y \subset X$ and $d' = d|_{Y \times Y}$, then (Y, d') is said to be a subspace of (X, d) .

* Defn 8.3

Let (X, d) be a metric space, $x \in X$, $\delta \in \mathbb{R}$ s.t. $\delta > 0$ then

$$\text{Open Ball / Ball} \quad B(x, \delta) := \{y \in X : d(x, y) < \delta\}$$

$$\text{Closed Ball} \quad C(x, \delta) := \{y \in X : d(x, y) \leq \delta\}$$

* Defn 8.4

Let (X, d) be a metric space.

* A subset $Y \subset X$ is called open if $\forall x \in Y \exists \delta > 0$ s.t. $B(x, \delta) \subset Y$

* A subset $Y \subset X$ is called closed if $X \setminus Y$ is open.

* If $x \in V$ and V is open then V is called an open neighbourhood of x .

* It is not necessary for a set to be either closed or open.

* Lemma 8.5

Let (X, d) be a metric space then

(i) $\emptyset$ and X are open subsets

(ii) If $V_1, V_2, \dots, V_k$ are open subsets then $\tilde{V} := \bigcap_{i=1}^k V_i$ is also open

(iii) If $\{V_\lambda\}_{\lambda \in I}$ are open subsets then $\tilde{V} := \bigcup_{\lambda \in I} V_\lambda$ is also open.

* Lemma 8.6

Let (X, d) be a metric space then

(i) $\emptyset$ and X are closed subsets

(ii) If $V_1, V_2, \dots, V_k$ are closed sets then $\tilde{V} := \bigcup_{i=1}^k V_i$ is a closed set

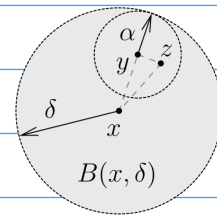
(iii) If $\{V_\lambda\}_{\lambda \in I}$ are closed then $\tilde{V} := \bigcap_{\lambda \in I} V_\lambda$ is a closed set

Proofing: Use De Morgan's law 1.5

* Lemma 8.7 Let (X, d) be a metric space, $x \in X$, $\delta \in \mathbb{R}$ s.t. $\delta > 0$ then

- (i) Open Ball $B(x, \delta)$ is an open subset of X
- (ii) Closed Ball $C(x, \delta)$ is a closed subset of X

Hint



* Theorem 8.8 Let (X, d) be a metric space, $Y \subset X$, and $V \subset Y$, then

$$V \text{ is open in } Y \iff \exists V \subset X \text{ s.t. } \begin{array}{l} \textcircled{i} V \text{ is open in } X \\ \textcircled{ii} V = V \cap Y \end{array}$$

Hint (i) $A \iff B$: if $x \in V$ then $B_Y(x, \delta) = B_X(x, \delta) \cap Y$

(ii) $A \implies B$: $V := \bigcup_{x \in V} B_X(x, \delta_x)$ where $B_Y(x, \delta_x) \subset V$

* Lemma 8.9 Let (X, d) be a metric space, $Y \subset X$ be open in X , $V \subset Y$

$$V \text{ is open in } X \iff V \text{ is open in } Y$$

Hint: 8.8, 8.5

* Lemma 8.10 Let (X, d) be a metric space, $Y \subset X$ be closed in X , $V \subset X$

$$V \text{ is closed in } X \iff V \text{ is closed in } Y$$

Hint: $A \iff B$, $\frac{Y}{V}$ is open in Y . Using 8.9 $\exists V_Y$ s.t. $Y/V_Y = V \cap Y \implies V = Y/V_Y = Y/V \cup \frac{Y}{V} = \frac{Y}{V}$

Then using 1.5.1 $\frac{X}{V} = \frac{X}{(Y/V)} = \frac{X}{Y} \cup (X/V_Y) = \frac{X}{Y} \cup \frac{X}{V}$ $\therefore$ using 8.5 $\frac{Y}{V}$ is open in X

* Def 8.11 A non empty metric space (X, d) is called connected if its only subsets that are both closed and open (clopen) are $\emptyset$ and X .

* Lemma 8.12 Let (X, d) be a connected metric space, $X_1, X_2 \subset X$ s.t.

- ① X_1 and X_2 are open in X
 - ② $X_1 \cup X_2 = X$ ③ $X_1 \cap X_2 = \emptyset$
- $\implies$
- either X_1 or X_2 is $\emptyset$ and the other one is X

* Lemma 8.13 Let (X, d) be a metric space, $S \subset X$

$$S \text{ is disconnected} \iff \exists U_1, U_2 \subset X \text{ s.t.}$$

$$\textcircled{i} U_1 \text{ and } U_2 \text{ are open in } X$$

$$\textcircled{ii} U_1 \cap U_2 \cap S = \emptyset$$

$$\textcircled{iii} (U_1 \cap S) \cup (U_2 \cap S) = S$$

$$\textcircled{iv} U_1 \cap S \neq \emptyset \text{ and } U_2 \cap S \neq \emptyset$$

* Defn 8.14 Let (X, d) be a metric space. Then the closure of $A \subset X$ is defined as

$$\bar{A} := \bigcap \{E \subset X : E \text{ is closed in } X \text{ and } A \subseteq E\}$$

* Lemma 8.15 Let (X, d) be a metric space. $A \subset X$, then

① $\bar{A}$ is a closed set

Hint 8.6

② $\bar{A} = A \iff A$ is a closed set

* Lemma 8.16 Let (X, d) be a metric space, $A \subset X$, then

$$x \in \bar{A} \iff \forall \delta > 0 \quad B_\delta(x, \delta) \cap A \neq \emptyset$$

Hint Prove by contrapositives $x \notin \bar{A} \iff \exists \delta > 0 \text{ st } B(x, \delta) \cap A = \emptyset$

$$x \notin \bar{A} \Rightarrow x \in A^c \subset \bar{A}^c \quad \begin{matrix} A \subset X \\ B(x, \delta) \end{matrix} \quad \bar{A}^c$$

* Defn 8.17 Let (X, d) be a metric space, $A \subset X$, then

the interior of A , i.e. $A^\circ := \{x \in A : \exists \delta > 0 \text{ st } B_\delta(x, \delta) \subset A\}$

boundary of A , i.e. $\partial A := \bar{A} \setminus A^\circ$

* Lemma 8.18 Let (X, d) be a metric space, $A \subset X$, then

① A° is an open set

Hint (i) $x \in A^\circ \Rightarrow \exists \delta > 0 \quad B(x, \delta) \subset A$. Let $z \in B(x, \delta)$
 open in X

② ∂A is a closed set

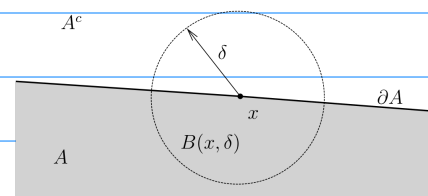
$$\Rightarrow \exists \epsilon > 0 \quad B(z, \epsilon) \subset B(x, \delta) \subset A \Rightarrow z \in A^\circ \Rightarrow B(x, \delta) \subset A^\circ$$

Hint (ii) Use 1.4 and 8.6

* Theorem 8.19 Let (X, d) be a metric space, $A \subset X$, then

$$x \in \partial A \iff \forall \delta > 0 \quad B(x, \delta) \cap A \neq \emptyset$$

$$B(x, \delta) \cap A^c \neq \emptyset$$



Hint $x \in \bar{A} \setminus A^\circ \implies x \notin A^\circ, \exists \delta > 0 \text{ st } B(x, \delta) \cap A^c \neq \emptyset \text{ then } B(x, \delta) \cap A \neq \emptyset \implies x \in \bar{A}$ contradiction

$$\implies x \in \bar{A} \xrightarrow{8.16} \forall \delta > 0 \quad B(x, \delta) \cap A \neq \emptyset$$

$$x \in \frac{A}{A^c} = \frac{A}{A} \cup (A \cap A^c) = \frac{A}{A} \cup \emptyset \implies x \in \frac{A}{A} \implies \exists \delta > 0 \quad B(x, \delta) \subset \frac{A}{A} \implies B(x, \delta) \cap A = \emptyset$$

$$\implies x \in \bar{A} \implies \exists \delta > 0 \quad B(x, \delta) \cap A \neq \emptyset \implies B(x, \delta) \cap A^c = \emptyset$$

Corollary 8.20

Let (X, d) be a metric space, $A \subset X \implies \partial A = \bar{A} \cap \bar{A}^c$

Hint 8.16, 8.19

Defn 8.21

A sequence $\{x_n\}_{n=1}^{\infty}$ of elements in the metric space (X, d) , is a function $f: \mathbb{N} \rightarrow X$

Defn 8.22

A sequence $\{x_n\}_{n=1}^{\infty}$ in the metric space (X, d) is called bounded if

$$\exists p \in X \text{ and } B \in \mathbb{R} \text{ st } d(x_n, p) \leq B \quad \forall n \in \mathbb{N}$$

Defn 8.23

A sequence $\{x_n\}_{n=1}^{\infty}$ in the metric space (X, d) is called convergent to $p \in X$

$$\text{ie } \lim_{n \rightarrow \infty} x_n = p \text{ if } \forall \varepsilon > 0 \exists M \in \mathbb{N} \text{ st } \forall n \geq M \quad d(x_n, p) < \varepsilon$$

Lemma 8.24

If the limit of $\{x_n\}_{n=1}^{\infty}$ in (X, d) exists then it is unique.

Hint: Similar to 3.5

Lemma 8.25

If $\{x_n\}_{n=1}^{\infty}$ in (X, d) is convergent then it is bounded

Hint: Similar to 3.6

Lemma 8.26

Let $\{x_n\}_{n=1}^{\infty}$ be a sequence in (X, d) , then

$$\lim_{n \rightarrow \infty} x_n = p \iff \exists \{a_n\}_{n=1}^{\infty} \subset \mathbb{R} \text{ st } d(x_n, p) < a_n \text{ and } \lim_{n \rightarrow \infty} a_n = 0$$

Lemma 8.27

Let $\{x_n\}_{n=1}^{\infty}$ be a sequence in (X, d) , then① If $\lim_{n \rightarrow \infty} x_n = p \implies$ All subsequences $\{x_{n_k}\}_{k=1}^{\infty}$ converge to p .② If for some $k \in \mathbb{N}$ $\{x_n\}_{n=k+1}^{\infty}$ is $\implies \{x_n\}_{n=1}^{\infty}$ is convergent to $p \in X$
convergent to $p \in X$

* Lemma 8.28

Let $\{x_n\}_{n=1}^{\infty}$ be a sequence in the euclidean space $\mathbb{R}^n$, where $x_n = (x_{n,1}, x_{n,2}, \dots, x_{n,n}) \in \mathbb{R}^n$, then

$$\{x_n\}_{n=1}^{\infty} \text{ converges} \iff \{x_{n,k}\}_{n=1}^{\infty} \text{ converges } \forall 1 \leq k \leq n$$

$$\text{Also } \lim_{n \rightarrow \infty} x_n = \left(\lim_{n \rightarrow \infty} x_{n,1}, \lim_{n \rightarrow \infty} x_{n,2}, \dots, \lim_{n \rightarrow \infty} x_{n,n} \right)$$

* Lemma 8.29
★

Let $\{x_n\}_{n=1}^{\infty}$ be a sequence in the metric space (X, d) , then

$$\lim_{n \rightarrow \infty} x_n = p \iff \forall \text{ open neighbourhoods } \mathcal{V} \text{ of } p, \exists M \in \mathbb{N} \text{ s.t.} \\ \forall n \geq M \quad x_n \in \mathcal{V}$$

* Lemma 8.30
★

Let (X, d) be a metric space, $E \subset X$

$$E \text{ is closed} \iff \forall \{x_n\}_{n=1}^{\infty} \text{ s.t. } \forall n \in \mathbb{N} \quad x_n \in E, \exists \lim_{n \rightarrow \infty} x_n = x \text{ then } x \in E$$

Hint: True $\Rightarrow$ True Proof by contradiction let $x \in X \setminus E$ Use 8.29

False $\Rightarrow$ False Let E is not closed, $x \in \bar{E} \setminus E$, Use 8.19 to $y_n \in B(x, \frac{1}{n}) \cap E$

* Lemma 8.31

Let (X, d) be a metric space, $A \subset X$, then

$$p \in \bar{A} \iff \exists \{x_n\}_{n=1}^{\infty} \text{ s.t. } x_n \in A \quad \forall n \in \mathbb{N} \text{ and } \lim_{n \rightarrow \infty} x_n = p$$

Use 8.16 and 8.30

* Defn 8.32
Cauchy

Let $\{x_n\}_{n=1}^{\infty}$ be a sequence in the metric space (X, d) . Then $\{x_n\}_{n=1}^{\infty}$ is called Cauchy if

$$\forall \varepsilon > 0 \quad \exists M \in \mathbb{N} \text{ s.t. } \forall n, k \geq M \quad d(x_n, x_k) < \varepsilon$$

* Lemma 8.33

If $\{x_n\}_{n=1}^{\infty}$ in (X, d) is convergent then it is Cauchy.

* Defn 8.34

A metric space (X, d) is called Cauchy complete if every Cauchy sequence in X is convergent to a point p in X

* Lemma 8.35

Finite dimensional space $\mathbb{R}^n$ with the standard metric is Cauchy complete.

Use 8.30 and 8.34

* Lemma 8.36

Let (X, d) be a complete metric space, $E \subset X$ be a closed set. Then E is Cauchy complete

* Defn 8.37

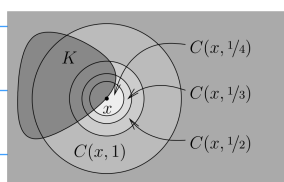
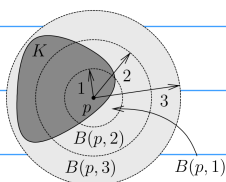
Let (X, d) be a metric space, $K \subset X$, then K is called compact if for every

$$K \subset \bigcup_{\lambda \in I} \mathcal{V}_{\lambda} \text{ where } \mathcal{V}_{\lambda} \text{ is an open set,}$$

there exists a finite subset $\{\lambda_1, \lambda_2, \dots, \lambda_j\}$ s.t.

$$K \subset \bigcup_{\lambda=1}^j \mathcal{V}_{\lambda}$$

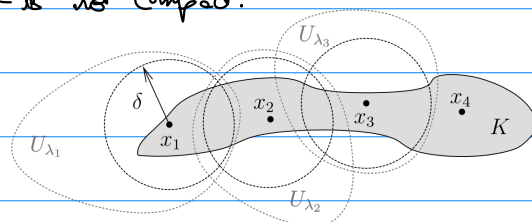
* Lemma 8.38 Let (X, d) be a metric space, if $K \subset X$ is compact then K is bounded and closed



Hint $\text{not closed} \Rightarrow \text{not compact}$
 $\bar{K} \neq K \quad x \in \bar{K} \setminus K$

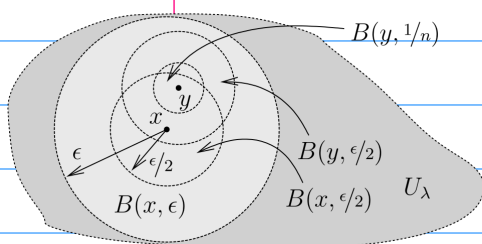
* Lemma 8.39 Let (X, d) be a metric space, $K \subset X$, $\{x_n\}_{n=1}^{\infty}$ a sequence in K s.t. $\forall x \in K$ and $\exists \alpha_x > 0$ $B(x, \alpha_x)$ contains finitely many points of $\{x_n\}_{n=1}^{\infty}$ then K is not compact.

Hint $K \subset \bigcup_{x \in K} B(x, \alpha_x)$. Then any finite cover of this open cover has only finitely many points of x_n , i.e. $\nexists$ a finite cover



* Defn 8.40 Let (X, d) be a metric space. Then $K \subset X$ is called sequentially compact if every sequence in K has a subsequence convergent to a point in K .

* Lebesgue 8.41 covering lemma Let (X, d) be a metric space, $K \subset X$ be sequentially compact, then given an open cover $\{U_{\lambda}\}_{\lambda \in I}$ of K , $\exists$ a $\delta > 0$ s.t. $\forall x \in K \quad B(x, \delta) \subset U_{\lambda}$ where $\lambda \in I$.



Proof by contradiction $\{(x_n, \lambda_n)\}_{n=1}^{\infty}$ s.t. $B(x_n, \lambda_n) \not\subset U_{\lambda} \quad \forall \lambda \in I$

Take $x \in K$ then $x \in U_{\lambda}$ $\exists \epsilon > 0$ s.t. $B(x, \epsilon) \subset U_{\lambda}$

Let $y \in B(x, \epsilon/2)$ then $B(y, \epsilon/2) \subset B(x, \epsilon) \subset U_{\lambda}$

$\forall n \geq 2/\epsilon \quad x_n \neq y$ because $B(x_n, \lambda_n) \not\subset U_{\lambda}$

$\therefore \lim_{k \rightarrow \infty} x_k \neq x$

* Theorem 8.42

Let (X, d) be a metric space, $K \subset X$

K is compact $\iff K$ is sequentially compact

Hint compact $\Rightarrow$ sequentially compact Use 8.39

sequentially compact $\Rightarrow$ compact

Let $K \subset \bigcup_{\lambda \in I} U_{\lambda}$ (Open cover)

Take $x_1 \in K$, Using Lebesgue covering lemma $\exists \delta > 0$ s.t. $B(x_1, \delta) \subset U_{\lambda_1}$ where $\lambda_1 \in I$

if $K \subset U_{\lambda_1}$ we are done otherwise take $x_2 \in K \setminus U_{\lambda_1}$ then $d(x_1, x_2) \geq \delta$ and $B(x_2, \delta) \subset U_{\lambda_2}$

if $K \subset U_{\lambda_1} \cup U_{\lambda_2}$ we are done or we continue the construction.

Either we end up a finite subcover or a sequence in K that doesn't have any convergent subsequences

* Corollary
8.43

Let (X, d) be a metric space. $K \subset X$ be compact,

if $E \subset K$ is closed $\implies E$ is compact.

Hint: Use 8.42 and 8.20

* Heine-Borel
Theorem 8.44

A closed and bounded set $K \subset \mathbb{R}^n$ is compact.

Hint: Use Bolzano Weierstrass theorem and 8.42

* Lemma
8.44(a)

Let (X, d) be a metric space, $K \subset X$

If K is compact $\implies K$ is Cauchy complete

Hint: Let $\{x_n\}_{n=1}^{\infty}$ be Cauchy, Using 8.42 $\{x_{n_k}\}_{k=1}^{\infty}$ s.t. $\lim_{k \rightarrow \infty} x_{n_k} = x_0$. Let $\epsilon > 0$

$\exists M_1 \in \mathbb{N} \forall n, k \geq M_1, d(x_n, x_k) < \epsilon/2$, $\exists M_2$ s.t. $\forall n_k \geq M_2, d(x_{n_k}, x_0) < \epsilon/2$

$n \geq \max(M_1, M_2) \implies d(x_n, x_0) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x_0) < \epsilon/2 + \epsilon/2 < \epsilon$

* Defn 8.45
Continuous
function

Let (X, d_X) and (Y, d_Y) be metric spaces, $f: X \rightarrow Y$ be a function, $c \in X$, then

f is called continuous at c if $\forall \epsilon > 0 \exists \delta > 0$ s.t. if $d_X(x, c) < \delta$ then

$d_Y(f(x), f(c)) < \epsilon$

* Lemma 8.46

Let (X, d_X) , (Y, d_Y) be metric spaces, $c \in X$, $f: X \rightarrow Y$ a function then

f is continuous at $c \iff \forall \{x_n\}_{n=1}^{\infty}$ s.t. $\lim_{n \rightarrow \infty} x_n = c$ then $\lim_{n \rightarrow \infty} f(x_n) = f(c)$

* Lemma 8.47

Let (X, d_X) , (Y, d_Y) be metric spaces and $f: X \rightarrow Y$ a continuous function.

If $K \subset X$ is a compact set, then $f(K)$ is also a compact set

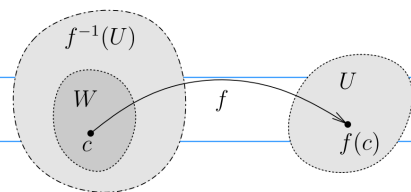
Hint: Use 8.42

* Extreme
Value Theorem 8.48

Let (X, d) be a non empty compact metric space and let $f: X \rightarrow \mathbb{R}$ be a continuous function.

Then f is bounded and achieves its absolute maxima and minima on X .

Hint 8.47, 8.42, 8.38



* Theorem 8.49

✱

Let $(X, d_X), (Y, d_Y)$ be metric spaces, $c \in X$, $f: X \rightarrow Y$, then
 f is continuous at $c \iff \forall$ open neighbourhoods $\mathcal{V}$ of $f(c)$ in Y ,
 $f^{-1}(\mathcal{V})$ contains an open neighbourhood of c in X

Hint: $A \Rightarrow B$ Let $\mathcal{V}$ be an open neighbourhood of $f(c) \rightarrow \exists \varepsilon > 0$ $B(f(c), \varepsilon) \subset \mathcal{V}$, from continuity $\exists \delta > 0$ s.t.
 if $d_X(x, c) < \delta$ then $d_Y(f(x), f(c)) < \varepsilon$ i.e. $f(B_X(x, \delta)) = B_Y(f(x), \varepsilon) \subset \mathcal{V}$

* Theorem 8.50

✱

Let $(X, d_X), (Y, d_Y)$ be metric spaces, $f: X \rightarrow Y$ be a function, then
 f is continuous on $X \iff \forall$ open sets $\mathcal{V} \subset Y$, $f^{-1}(\mathcal{V})$ is open in X

Hint f is continuous $\implies \forall \mathcal{V} \subset Y$ that is open, $f^{-1}(\mathcal{V})$ is open
 $A = f^{-1}(\mathcal{V}) = \{x \in X : f(x) \in \mathcal{V}\} \xleftrightarrow{\text{proj is conf. set}} f^{-1}(\mathcal{V}) = \emptyset$

Let $c \in A \implies f(c) \in \mathcal{V} \implies \exists \varepsilon > 0$ $B_Y(f(c), \varepsilon) \subset \mathcal{V} \implies f^{-1}(B_Y(f(c), \varepsilon)) \subset A \implies$ As f is continuous at c (8.49)
 A is open $\iff \therefore B_X(c, \delta) \subset f^{-1}(\mathcal{V}) \iff$ if $y \in B_Y(f(c), \varepsilon)$ then $f(x) \in B_Y(f(c), \varepsilon) \subset \mathcal{V} \iff \exists \delta > 0$ s.t. if $d_X(x, c) < \delta$ $d_Y(f(x), f(c)) < \varepsilon$

* Defn 8.51

Let $(X, d_X), (Y, d_Y)$ be metric spaces, then $f: X \rightarrow Y$ is called uniformly continuous
 if $\forall \varepsilon > 0 \exists \delta > 0$ s.t. if $p, q \in X$ and $d_X(p, q) < \delta$ then $d_Y(f(p), f(q)) < \varepsilon$.

* Theorem 8.52

✱

Let $(X, d_X), (Y, d_Y)$ be metric spaces, X a compact set and $f: X \rightarrow Y$ a continuous
 function then f is uniformly continuous

Hint f because of continuity
 Let $\varepsilon > 0$, $\forall c \in X$ find δ_c s.t. if $x \in B(c, \delta_c) \implies d(f(x), f(c)) < \varepsilon/2$

$X \subset \bigcup_{c \in X} B(c, \delta_c)$ is an open cover. X is compact $\therefore$ using Lebesgue's lemma $\exists \underline{\delta} > 0$ s.t.
 if $x \in X$ $B(x, \underline{\delta}) \subset B(c, \delta_c)$ for some (c, δ_c) of open cover

Let $p, q \in X$ s.t. $d(p, q) < \underline{\delta}$, Using Lebesgue find (c, δ_c) s.t. $B(p, \underline{\delta}) \subset B(c, \delta_c)$
 $d(f(p), f(q)) \leq d(f(p), f(c)) + d(f(q), f(c)) < \varepsilon_1 + \varepsilon_2 < \varepsilon$

* Corollary 8.53

If $f: [a, b] \times [c, d] \rightarrow \mathbb{R}$ is a continuous function, then $g: [c, d] \rightarrow \mathbb{R}$ defined by

$$g(x) := \int_a^b f(x, y) dy \text{ is uniformly continuous}$$

Hint: 8.52

find $\delta > 0$ s.t. $|x - y| < \delta \implies |f(x, z) - f(y, z)| < \varepsilon/6$
 $|g(x) - g(y)| < \varepsilon$

* Defn 8.54
Lipschitz
continuous

Let $(X, d_X), (Y, d_Y)$ be metric spaces, $f: X \rightarrow Y$ be a function. Then f is called Lipschitz continuous if $\exists K \in \mathbb{R}$ s.t. if $p, q \in X$ then

$$d_Y(f(p), f(q)) \leq K d_X(p, q)$$

* Lemma 8.55

Let $(X, d_X), (Y, d_Y)$ be metric spaces, $f: X \rightarrow Y$ be a function
 f is Lipschitz continuous $\implies f$ is uniformly continuous

* Defn 8.56
Cluster
Point

Let (X, d) be a metric space, $S \subset X$. Then $p \in S$ is called a cluster point if $\forall \varepsilon > 0$ $B(p, \varepsilon) \cap S \setminus \{p\} \neq \emptyset$

* Lemma 8.57

Let (X, d) be a metric space, $S \subset X$, $p \in S$, then

p is a cluster point of $S \iff \exists \{x_n\}_{n=1}^{\infty}$ s.t. $\forall n \in \mathbb{N} \ x_n \in S \setminus \{p\}$ and $\lim_{n \rightarrow \infty} x_n = p$

* Defn 8.58
Limit

Let (X, d) be a metric space, $c \in S$ a cluster point of $S \subset X$, $L \in Y$, $f: X \rightarrow Y$ a function.
 Then we say $\lim_{x \rightarrow c} f(x) = L$ if $\forall \varepsilon > 0 \ \exists \delta > 0$ s.t. if $x \in S \setminus \{c\}$ and $d_X(x, c) < \delta$ then $d_Y(f(x), L) < \varepsilon$

* Lemma 8.59

Let (X, d) be a metric space, $S \subset X$, $c \in S$ a cluster point of S , $f: X \rightarrow Y$ a function.

If $\exists \lim_{x \rightarrow c} f(x) = L \implies$ The limit is unique

* Lemma 8.60

Let (X, d) be a metric space, $c \in S$ a cluster point of $S \subset X$, $f: X \rightarrow Y$ a function.

Sequential limit

$\lim_{x \rightarrow c} f(x) = L \iff \forall \{x_n\}_{n=1}^{\infty}$ s.t. $\forall n \in \mathbb{N} \ x_n \in S \setminus \{c\}$, $\lim_{n \rightarrow \infty} x_n = c$ then $\lim_{n \rightarrow \infty} f(x_n) = L$

* Defn 8.61
Contraction
Mapping

Let $(X, d_X), (Y, d_Y)$ be metric spaces. Then a map $\phi: X \rightarrow Y$ is called a contraction

if ϕ is K -Lipschitz continuous and $K < 1$, i.e. if $p, q \in X$ then

$$d_Y(\phi(p), \phi(q)) \leq K d_X(p, q) \text{ and } K < 1$$

* Defn 8.62
Fixed Point

Let (X, d) be a metric space, $\phi: X \rightarrow X$, then $p \in X$ is called a fixed point if $\phi(p) = p$

* Banach
Fixed Point
Theorem 8.66

Let (X, d) be a nonempty complete metric space, $\phi: X \rightarrow X$ be a contraction mapping.
Then ϕ has a unique fixed point.

Hint: Let $x_0 \in X$, $x_{n+1} = \phi(x_n)$, $\{x_n\}_{n=1}^{\infty}$

$$d_n(x_{n+1}, x_n) = d_n(\phi(x_n), \phi(x_{n-1})) \leq K d_n(x_n, x_{n-1}) \dots \leq K^n d_n(x_1, x_0)$$

$$\text{Let } m, n \in \mathbb{N} \text{ s.t. } m > n \text{ then } d_n(x^m, x^n) \leq \sum_{i=n}^{m-1} d_n(x_{i+1}, x_i) \leq \sum_{i=n}^{m-1} K^i d_n(x_1, x_0) \leq K^n \frac{d_n(x_1, x_0)}{1-K}$$

$$\therefore \{x_n\}_{n=1}^{\infty} \text{ is Cauchy } \lim_{n \rightarrow \infty} x_n = x$$

$$\text{Then } \phi(x) = \phi\left(\lim_{n \rightarrow \infty} x_n\right) = \lim_{n \rightarrow \infty} \phi(x_n) = \lim_{n \rightarrow \infty} x_{n+1} = x \quad \therefore x \text{ is a fixed point}$$

$$\text{Let } x, y \text{ are two fixed points then } d(x, y) = d(\phi(x), \phi(y)) \leq K d(x, y) \Rightarrow x = y$$

* Lemma
8.64

Let $C([a, b], \mathbb{R})$ be the set of continuous functions $f: [a, b] \rightarrow \mathbb{R}$, if $g, h \in C([a, b], \mathbb{R})$
let $d_c := \|g - h\|_{[a, b]}$, then (C, d_c) is a complete metric space.

Proof: Let $\{f_n\}_{n=1}^{\infty}$ be a sequence that is Cauchy in uniform norm.

Then using 7.6, $\{f_n\}_{n=1}^{\infty}$ converges uniformly to $f: [a, b] \rightarrow \mathbb{R}$

Since f_n are all continuous functions and convergence is uniform, then using 7.7

$f: [a, b] \rightarrow \mathbb{R}$ is continuous and hence an element of C .

* Lemma
8.65

Let $J \subset \mathbb{R}$ be a closed interval, $\gamma = \{f \in C([a, b], \mathbb{R}) : f([a, b]) \subset J\}$, if $g, h \in \gamma$
 $d_c: \|g - h\|_{[a, b]}$, then (γ, d_c) is complete space

Proof: Let $\{f_n\}_{n=1}^{\infty}$ be a convergent sequence s.t. $\forall n \in \mathbb{N} \quad f_n \in \gamma$

since C is complete (8.64), $\lim_{n \rightarrow \infty} f_n = f \in C$

Let $x \in [a, b]$ $\{f_n(x)\}_{n=1}^{\infty}$ is a convergent sequence s.t. $\forall n \in \mathbb{N} \quad f_n(x) \in J$

Since $J \subset \mathbb{R}$ is closed $\lim_{n \rightarrow \infty} f_n(x) = f(x) \in J \Rightarrow f \in \gamma$

Using 8.30 $\gamma \subset C$ is closed and $\therefore$ Cauchy complete

* Picard's
Theorem

8.66

Let $I, J \subset \mathbb{R}$ be bounded and closed intervals, $(x_0, y_0) \in I_0 \times J_0$ where I_0, J_0 are interiors of I and J .

$F: I \times J \rightarrow \mathbb{R}$ s.t. F is continuous and Lipschitz continuous in second variable i.e.

$\exists L \in \mathbb{R}$ s.t. $\forall y, z \in J, x \in I \quad |F(x, y) - F(x, z)| \leq L|y - z|$, then

$\exists h > 0$ s.t. $[x_0 - h, x_0 + h] \subset I$ and there exists a unique differentiable function f s.t.

$f: [x_0 - h, x_0 + h] \rightarrow J \subset \mathbb{R}$, $f'(x) = F(x, f(x))$ and $f(x_0) = y_0$.

Proof: I, J are compact, F is continuous $\implies F$ is bounded $\implies \exists M \in \mathbb{R} \quad |f(x, y)| \leq M \quad \forall x \in I, y \in J$

let $x_0 = 0$. Find $\alpha > 0$ s.t. $[-\alpha, \alpha] \subset I$, $[y_0 - \alpha, y_0 + \alpha] \subset J$. $h := \min(\alpha, \frac{\alpha}{M+L})$

$\gamma := \{f \in C[-h, h] : f([-h, h]) \subset J\}$. $d_\gamma(f, g) = \|f - g\|_{[-h, h]}$

Using 8.64 and 8.65 (γ, d_γ) is a complete metric space

Let T be the mapping $T: \gamma \rightarrow \gamma$ $T(f)(x) := y_0 + \int_0^x F(t, f(t)) dt$

$|F(x, f(x)) - F(x, g(x))| \leq L|f(x) - g(x)| \leq L d_\gamma(f, g)$

$|T(f)(x) - T(g)(x)| \leq \left| \int_0^x (F(t, f(t)) - F(t, g(t))) dt \right| \leq \frac{Lx}{M+L} d_\gamma(f, g)$ } $\text{Prove } T \text{ is a contraction map}$

$d_\gamma(T(f)(x), T(g)(x)) \leq \frac{L}{M+L} d_\gamma(f, g)$

Further $|T(f)(x) - y_0| \leq \left| \int_0^x F(t, f(t)) dt \right| \leq \frac{Mx}{M+L} \leq \alpha$

Using fixed point theorem $f: [-h, h] \rightarrow [y_0 - \alpha, y_0 + \alpha]$ is the unique fixed point of T

i.e.

$f(x) = y_0 + \int_0^x F(t, f(t)) dt$

Hence proved